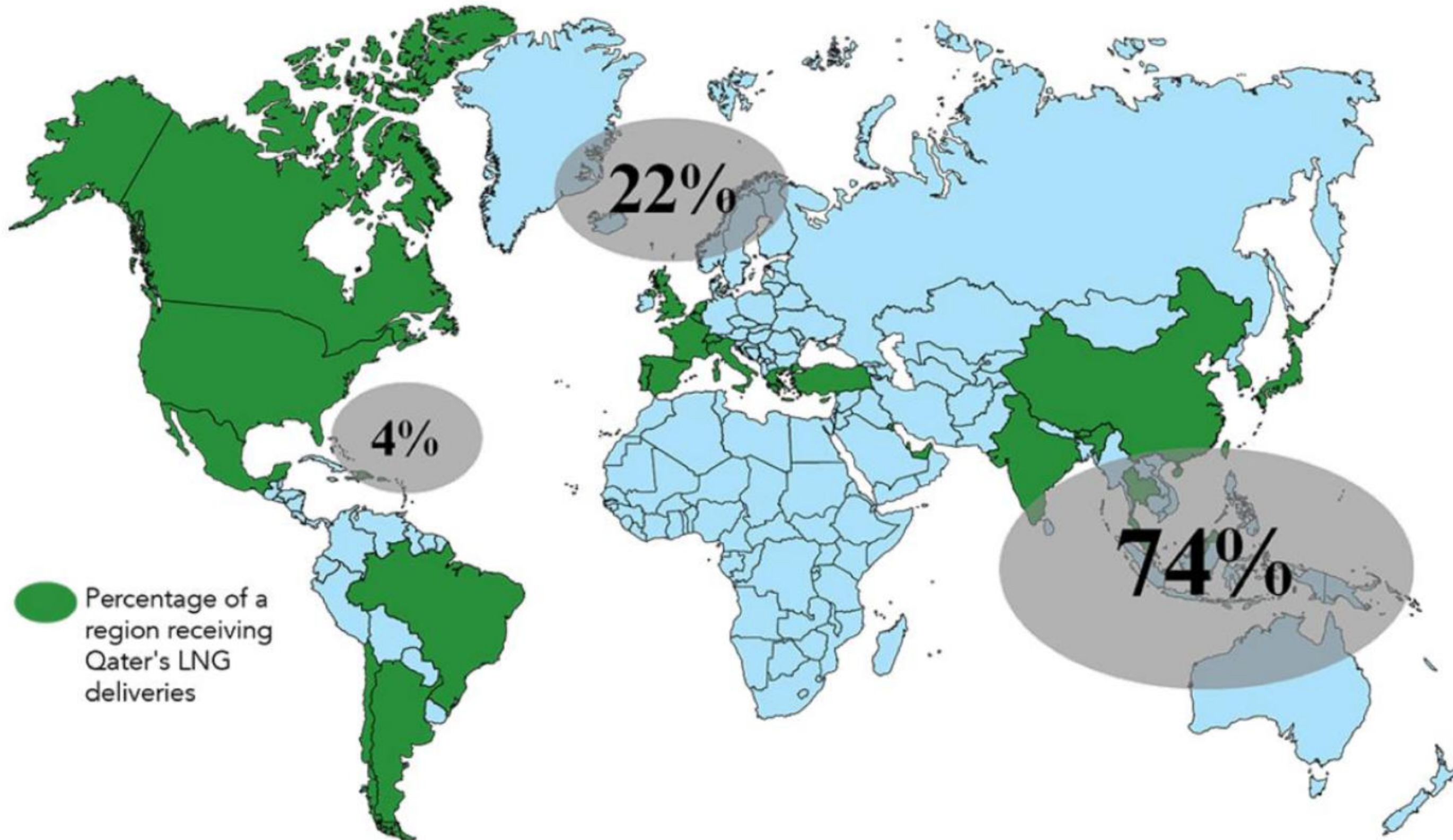


10.8 Detailed specific example: Qatar



The State of Qatar is a country in the Middle East. It occupies the north-eastern coast of the Arabian Peninsula and borders Saudi Arabia in the south. The capital is Doha, home to over 80% of the country's population. The country is a high-income economy, with the fourth-highest GDP per capita in the world. Qatar has the world's third largest reserves of natural gas and oil and is the world's largest exporter of liquefied natural gas. With such a large non-renewable energy source, the country has good energy security with 100% of the population having access to electricity. However, with long-term global climate stability becoming so important, there has been an increased interest in focusing on reducing reliance on fossil fuels.



Source **A**: Supply of liquefied natural gas to global markets from Qatar in 2023. Qatar supplies 33% of the world's LNG.

TASK 1: Study Source **A**

- Which region/country** receives the second largest supply of natural gas (LNG) from Qatar?
- Calculate the difference** between the region that receives the largest percentage of LNG from Qatar and the region that receives the lowest.
- Suggest how the export of large volumes of LNG to other parts of the world helps to support the Qatari economy.

The impacts of different energy sources

All energy sources have both positive and negative impacts, as discussed in Topic 10.6 and 10.7:

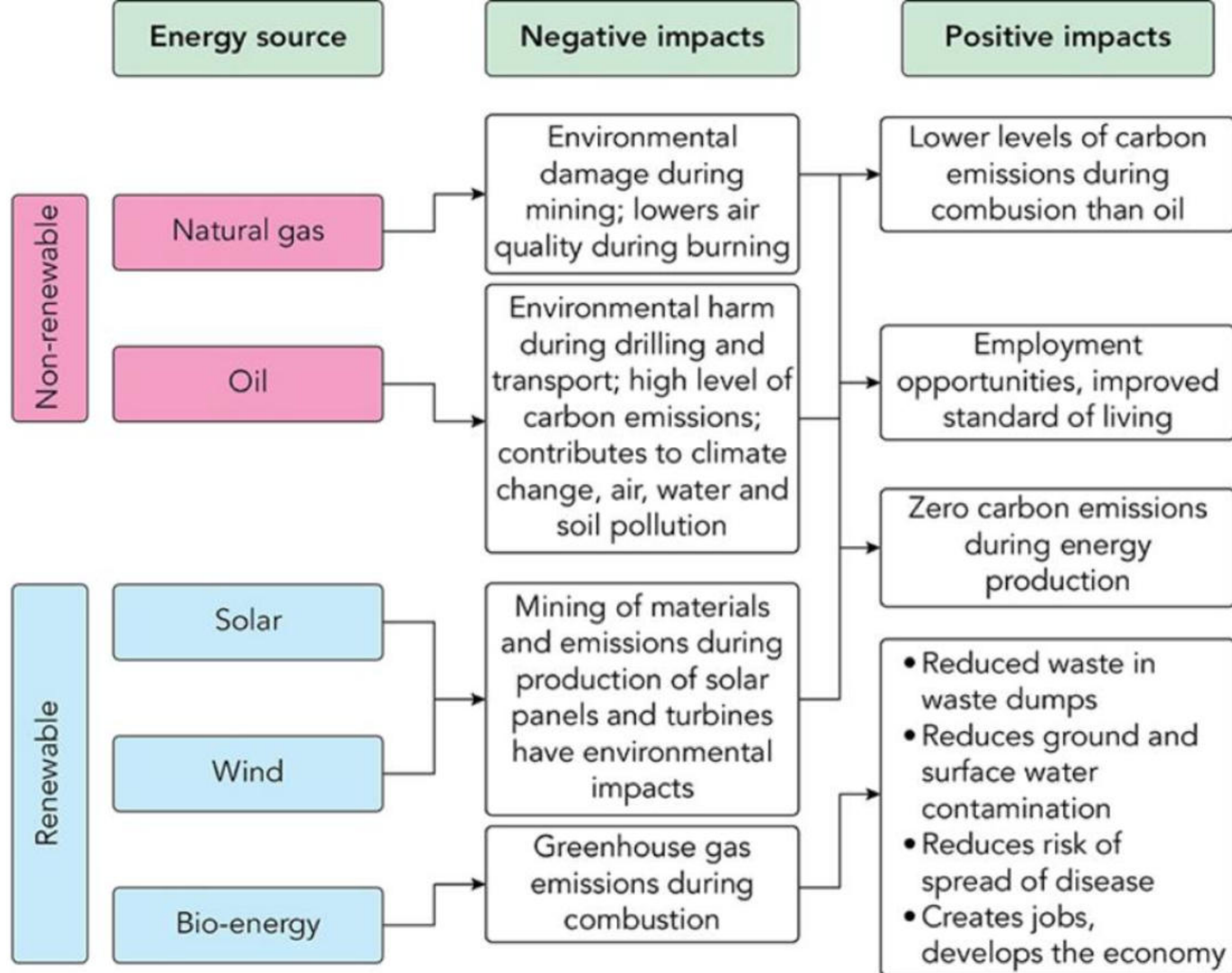


Figure 10.8: The impacts of different energy sources

Qatar's energy mix and strategies for managing energy supply

Qatar has historically relied on fossil fuels as an energy source due to its large reserves of natural gas and oil. However, in more recent years, Qatar has been developing a policy that will move them away from a reliance on fossil fuels to a wider range of energy sources. The long-term strategy aims to have 20% of their electricity supplied by solar energy by 2030.

Source B: The change in energy mix in Qatar between 2010 and 2022 in terawatt-hours (TWh)

Year	Oil (TWh)	Natural gas (TWh)	Solar (TWh)	Other renewables (TWh)
2010	79.81	254.06	0	0
2013	127.64	352.79	0.01	0.24
2022	154.88	366.96	1.07	0.43

TASK 2: Study Source B

- a** Which year had the highest energy production by natural gas?
- b** Calculate the:
 - i** difference in energy produced by natural gas between 2010 and 2022
 - ii** percentage change in energy produced by natural gas between 2010 and 2022
- c** Describe the change in energy produced by solar power between 2010 and 2022.
- d** Using the information from the detailed specific example, explain why the change observed in energy produced by solar energy occurred.

Qatar currently has a high economic dependency on oil and natural gas. The Qatari government has recently developed a number of strategies to reduce its dependence on oil. This has stemmed from an increased awareness of the links between fossil fuels and climate change, the instability of oil prices, and the increased development of global renewable energy technologies, which put the Qatari economy at risk due to other countries' potential decreased reliance on oil.

Reducing greenhouse gases

In August 2021, Qatar Energy (Qatar's national oil company) announced its intent to reduce greenhouse gas emissions by 25% by 2030. They have allocated \$170 million towards this effort. Qatar is also working towards developing electric transport options. To help support this goal, a plan is in place to install 1000 electric car charging stations by 2025. This will reduce harmful emissions that current fossil fuel reliant transport causes.

Energy efficiency

In 2021, Qatar's National Program for Conservation and Energy Efficiency saved the country approximately \$1.1 billion. This initiative aims to minimise the amount of wasted energy and increase the amount of renewable energy used. For example, through regulations, the programme has focused on techniques that ensure equipment that uses energy is doing so with minimal or no wasted energy. Smart meters also play an important role in the efforts to reduce energy consumption. Customers are able to monitor their usage and reduce it as they see consumption increasing.

Solar energy

Qatar has set the strategic goal of attaining 20% of its energy from solar power by 2030. Its desert climate with high rates of sunshine all year round, coupled with flat land and the financial means to install solar fields, means that it is possible for Qatar to benefit from concentrated solar power.

Manufacturing company Qatar Solar Technologies built a solar plant in 2017, with a total capacity of 1.1 megawatt hours (MWh). This energy source can power approximately 20 000 homes. In 2022, companies Total and Marubeni built a solar plant that produces 800 MWhs of electricity from 1.8 million solar panels. Two further solar power plants, which will generate 880 MWhs of electricity, are expected to be operational by the end of 2024.

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Energy from waste

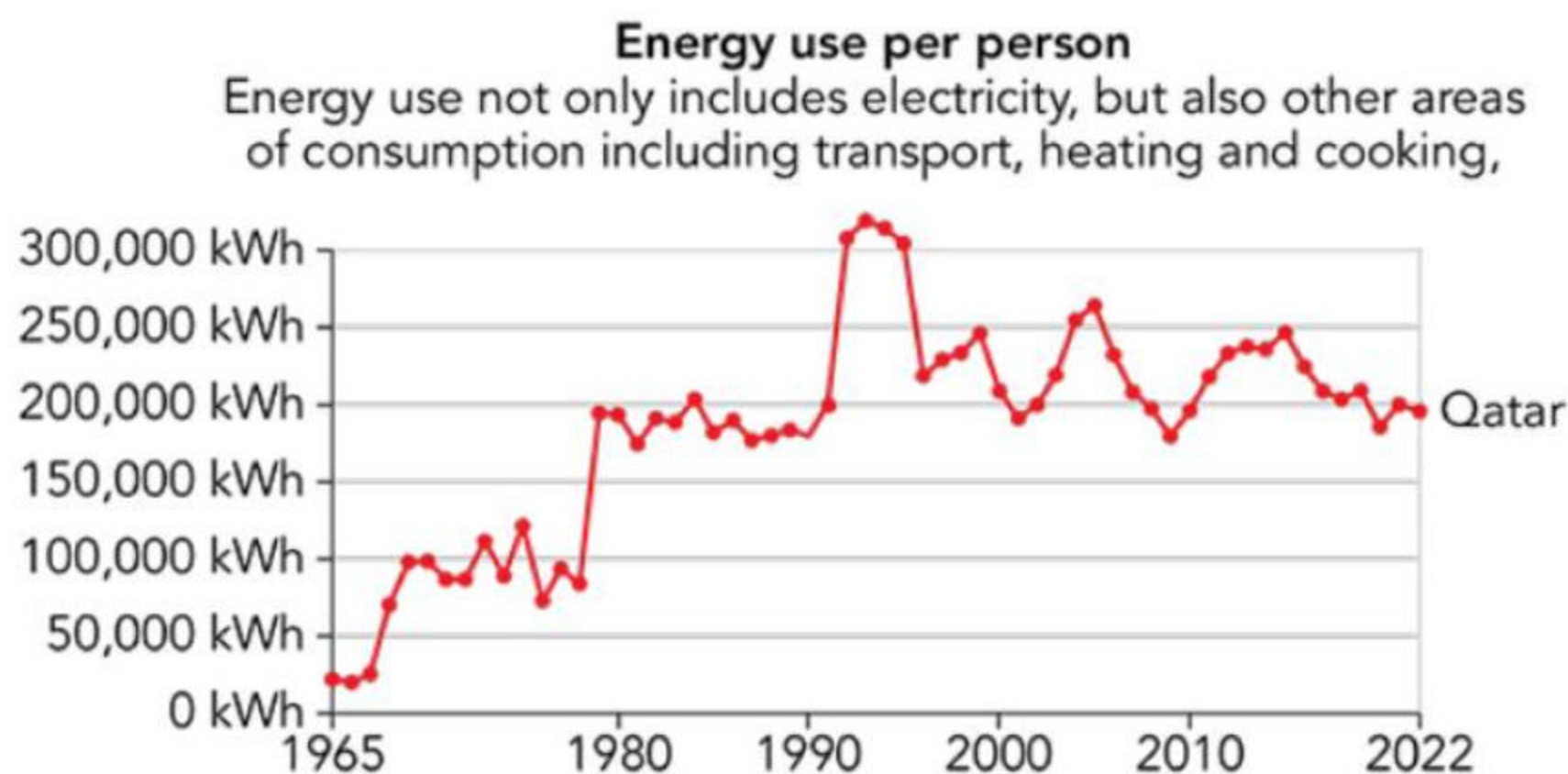
Waste-to-energy is where waste is turned into energy, such as electricity, heat, or transport fuels. The most common way of achieving this is through burning waste. The waste is burned at high temperatures and the heat is used to boil water. This creates steam that turns turbines, generating electricity. This energy source has two sustainability benefits: it reduces the waste going to landfill and it generates energy.

In Qatar, it is estimated that each individual generates 2 kg of waste every day, so this can all be used to generate power. Qatar was the first Gulf country to implement a waste-to-energy programme, generating over 50 MWh of electricity every day from its Domestic Solid Waste Management Centre. They have set their target at producing 100 MWh–150 MWh of electricity from waste every day by 2040.

Wind energy

Qatar has the potential for wind energy along the planned Qatar-Bahrain Causeway, where wind speeds are moderate and suited to small wind turbine generators for remote locations. Qatar is also exploring the potential of large-scale wind farm projects in the country. It has investigated the setup of a wind farm project with significant capacity in the north of the country.

To date, the strategies and techniques that Qatar has implemented have shown success in that they have developed and continue to develop a better energy mix. They have successfully used ways to reduce energy waste, and to develop functioning solar and energy-from-waste plants to supply clean energy to homes. With their goals set as far out as 2040, the success of long-term strategies are yet to be determined. However, the changes made to date indicate that they will continue to work towards fulfilling their long-term plans.



Source **C**: Energy used per capita in Qatar between 1965 and 2022

REFLECTION

Did you manage to connect the information found in Source C with what you've learned about Qatar's longterm energy security and energy mix plans? What do you think you can do to improve your ability to connect information learned in one place with information learned in another? Think about using flow diagrams, summaries or discussing the information with a classmate to look for connections.

- a** State which years had the highest energy use per person in Qatar.
- b** Describe the change in energy usage per person in Qatar between 1965 and 2022.
- c** In pairs, suggest three reasons for the changes you observed in energy usage per person over time in Qatar between 1965 and 2022.

IMPROVE THIS ANSWER

A country in Europe plans to increase their energy supply to meet future demand. They have three different plans:

- a** Build a new coal power station.
- b** Import large amounts of gas from the Middle East.
- c** Increase the amount of wind generators offshore and in the mountains.

Choose plan a, b or c that you think will be the most sustainable in the future. Explain your choice referring to the plan you have selected and the plans you have decided against.

Plan chosen: c Increase the amount of wind generators offshore and in the mountains.

The use of renewable energy has a number of advantages in the long term 1. These include the fact that renewable energy does not release carbon emissions into the atmosphere, helping in the fight against climate change due to increasing greenhouse gases. 2 Running costs will be low once the wind turbines are installed. 3 However, 4 the costs of producing and installing all the turbines will be high and will require investment. 5 In addition, some people may object to the wind turbines because they can be ugly and make a noise.

6 Building more coal power stations is a bad idea because of the air pollution that causes illness and death in the population, reducing the ability of people to work and slowing the economy. 7

8 Therefore, looking at the other two options, wind energy is a better option for long-term energy supply.

1 The answer refers to the long-term production of energy that the question is asking to be discussed.

2 and 3 The answer talks about a positive impact of the wind turbines.

4 The use of a word like 'however', 'in contrast', or 'in comparison to' indicate to the reader that you are thinking about a counter argument which is important when developing an evaluation.

5 The answer discusses the negative points of wind turbines, showing that both sides of the argument have been thought about.

6 The answer has discussed plan one but has only discussed the negatives. It needs to look for the counter argument to balance the argument—for example, the cost of energy due to possibly already having coal power stations.

7 The answer has not discussed the second plan, which is importing natural gas from the Middle East. The question asks for reference to both of the other plans. The answer needs to develop this, considering both the positive and negative impacts.

8 The conclusion needs to be developed more fully once the answer has discussed both the positive and negative impacts of both of the other plans.

Now, write an improved response to this answer using the guidance above.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics.
This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
describe how our food and energy is produced			
outline global patterns of food and energy supply and demand			
explain the reasons for and the impacts of the globalisation of food and energy supplies			
describe the role of food aid and the sustainable management of food supply in failing soils			
give the impacts of energy production from different energy sources			
evaluate the sustainable management of energy supplies			



› Theme 2

Practice questions

> Topic 6 Population change

1

- a Study Figure 1.1, which shows information about birth rates and death rates in Japan (a HIC) between 1970 and 2020.

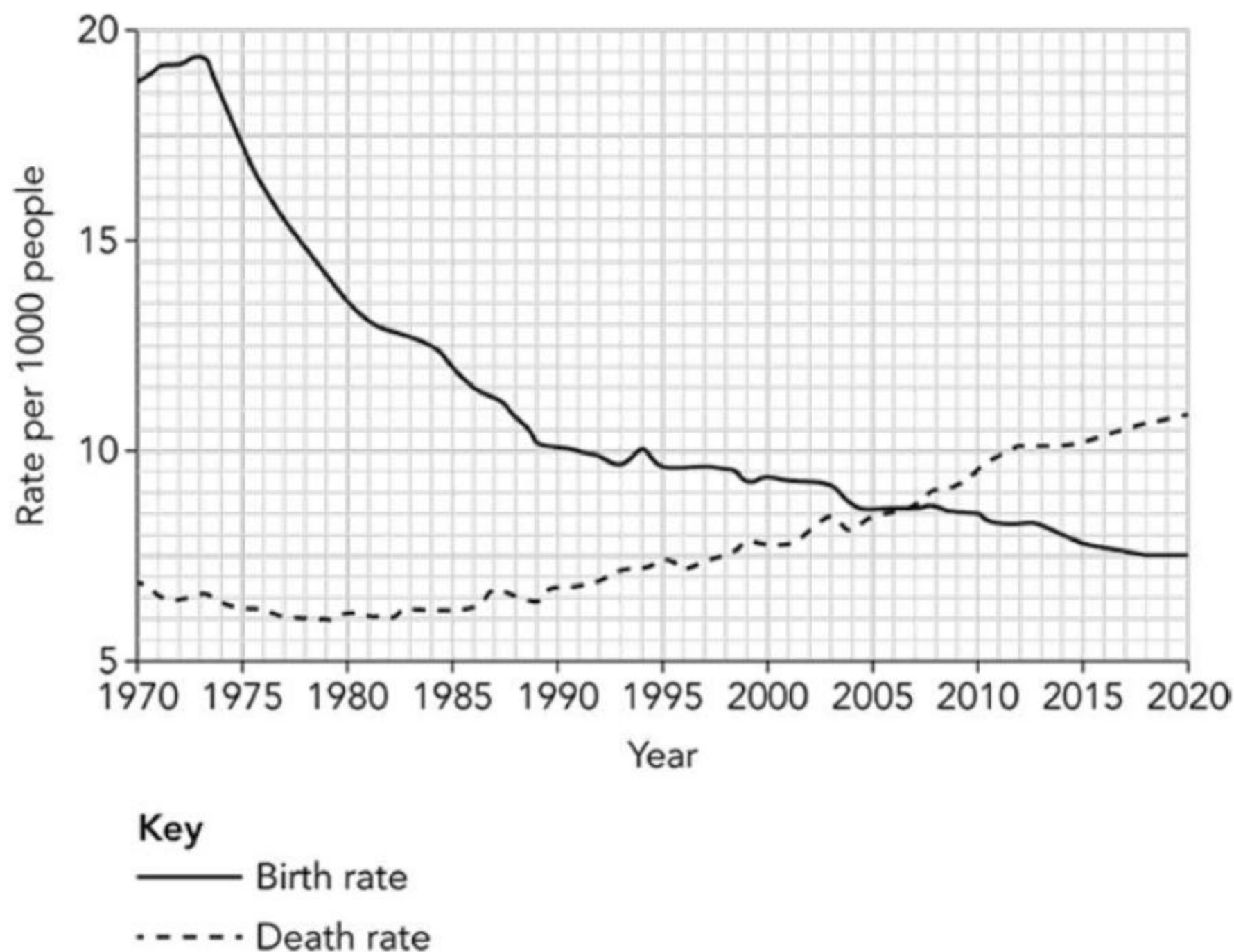


Figure 1.1

► Image Description

- i What was the birth rate, per 1,000, in Japan in 1970? [1]
 ii Which of the following statements are true about birth rates and death rates in Japan between 1970 and 2020?

Tick **two** answers in the table below.

[2]

	tick (✓)
The birth rate and the death rate remained constant.	
The birth rate decreased overall.	
The birth rate was always higher than the death rate.	
The death rate fluctuated more than birth rate.	
The death rate increased overall.	

- iii Calculate the rate of natural population decrease in Japan in 2020. You should show your calculations. [3]

Parts 1a – 1a iii: Adapted from Cambridge IGCSE Geography (0460)

Paper 12, Q1a, March 2023

- i Suggest three reasons why birth rates vary between countries. [3]

- b Study Figure 1.2, which shows the population structure for two countries, A and B.

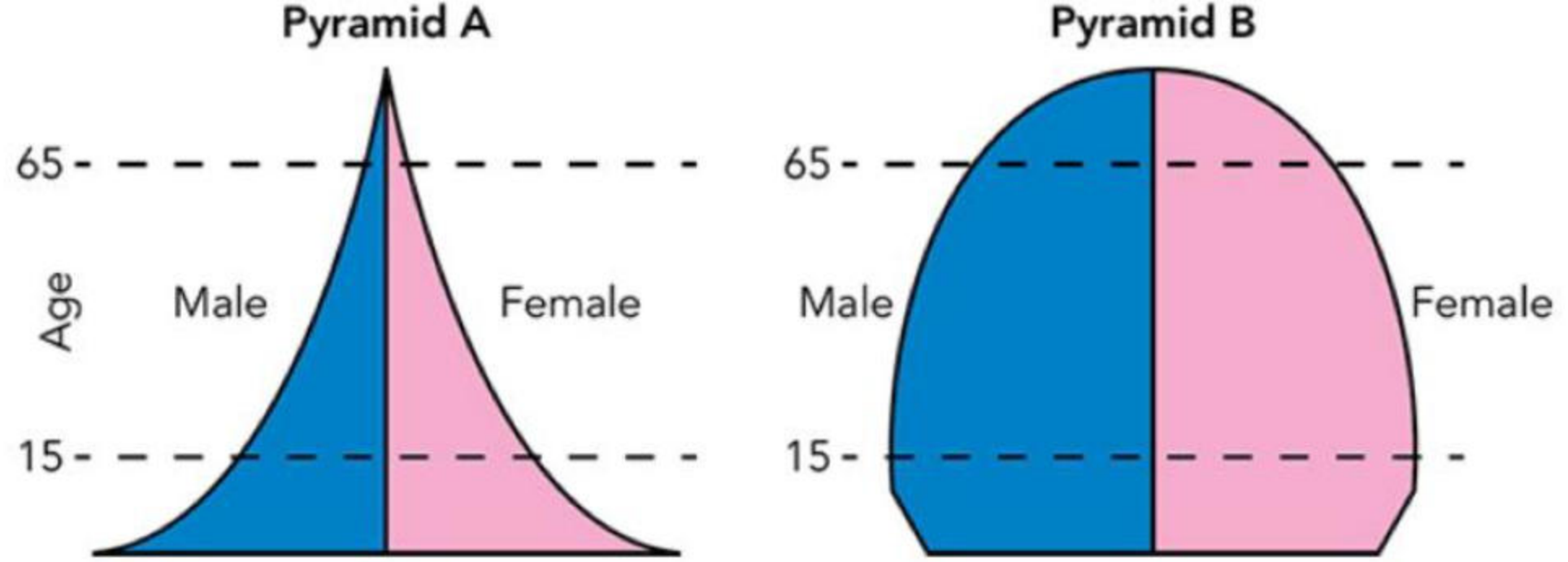


Figure 1.2

- i Which population pyramid shows the population structure of a HIC? Give three pieces of evidence to support your answer. [3]
- ii Describe and explain the positive and negative impacts for a country with an ageing population. [6]
- c For a named country you have studied, evaluate the positive and negative impacts of a government population policy aimed to encourage population growth. [7]

Parts 1a iv. – 1c: Please note that these parts of this question are not derived from the Cambridge IGCSE Geography (0460) past paper referenced above and were written by the author.

> Topic 7 Changing towns and cities

2

- a Study Figure 2.1, which shows the urban growth of HICs and LICs during the period 1950–2050.

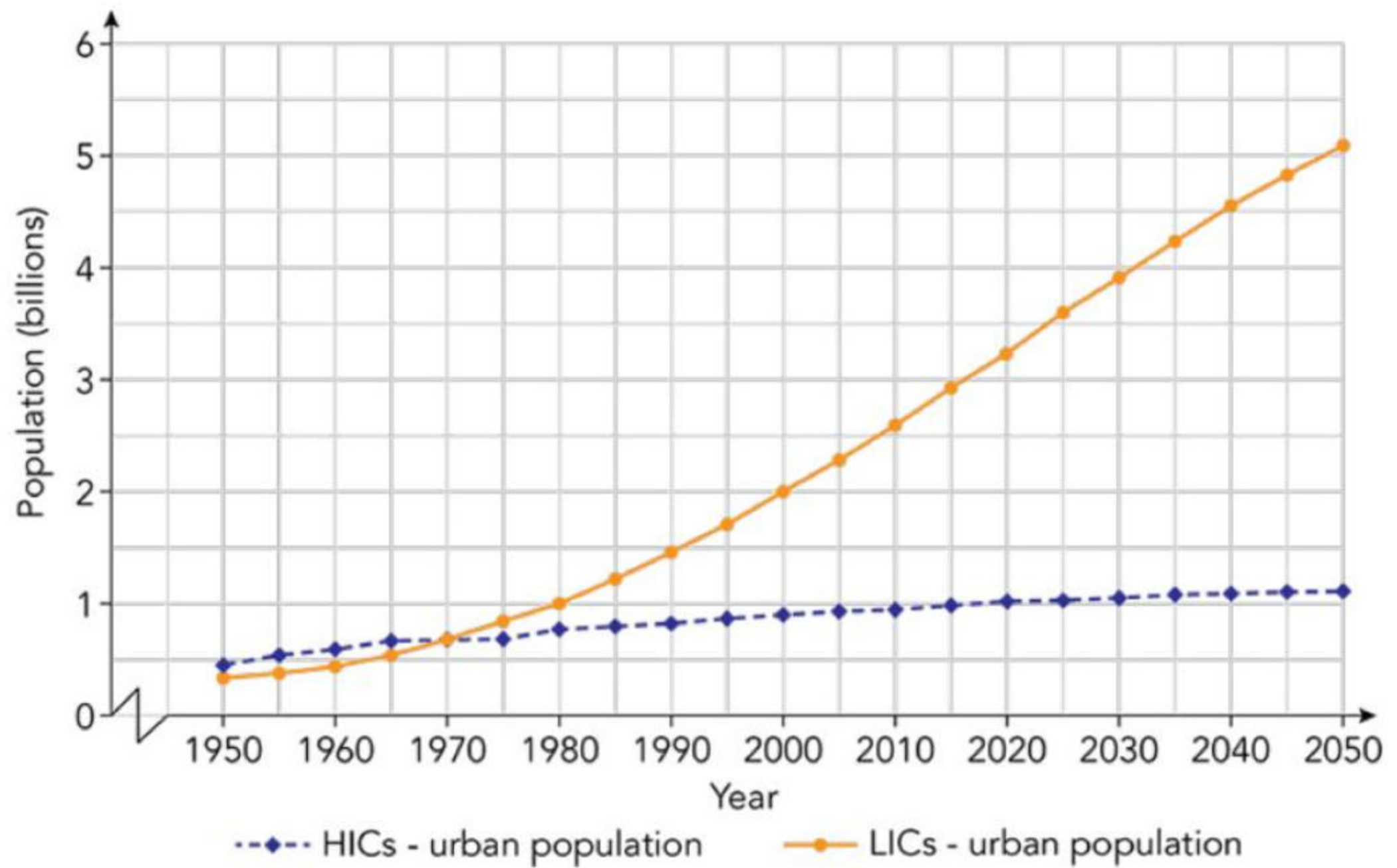


Figure 2.1

► Image Description

- | | | |
|-----|--|-----|
| i | Define the term 'urbanisation'. | [1] |
| ii | Using information from Figure 2.1, compare the urban growth of HICs and LICs since 2000. | [2] |
| iii | Suggest reasons for the rapid rates of urban growth in LICs. | [3] |
| iv | Explain two socio-economic challenges of rapid urban growth. | [4] |

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- b Study Figure 2.2, which shows a man repairing shoes on the streets of Delhi, India.



- i Suggest reasons why many people in cities in LICs work in the informal sector. [3]
- ii Describe and explain how urban areas may be made more sustainable. [5]
- c Evaluate the strategies and techniques used to sustainably manage the growth of urban areas. [7]

Author-written question

> Topic 8 The challenge of development

3

- a Study Figure 3.1, which shows the infant mortality rate in the Middle East.

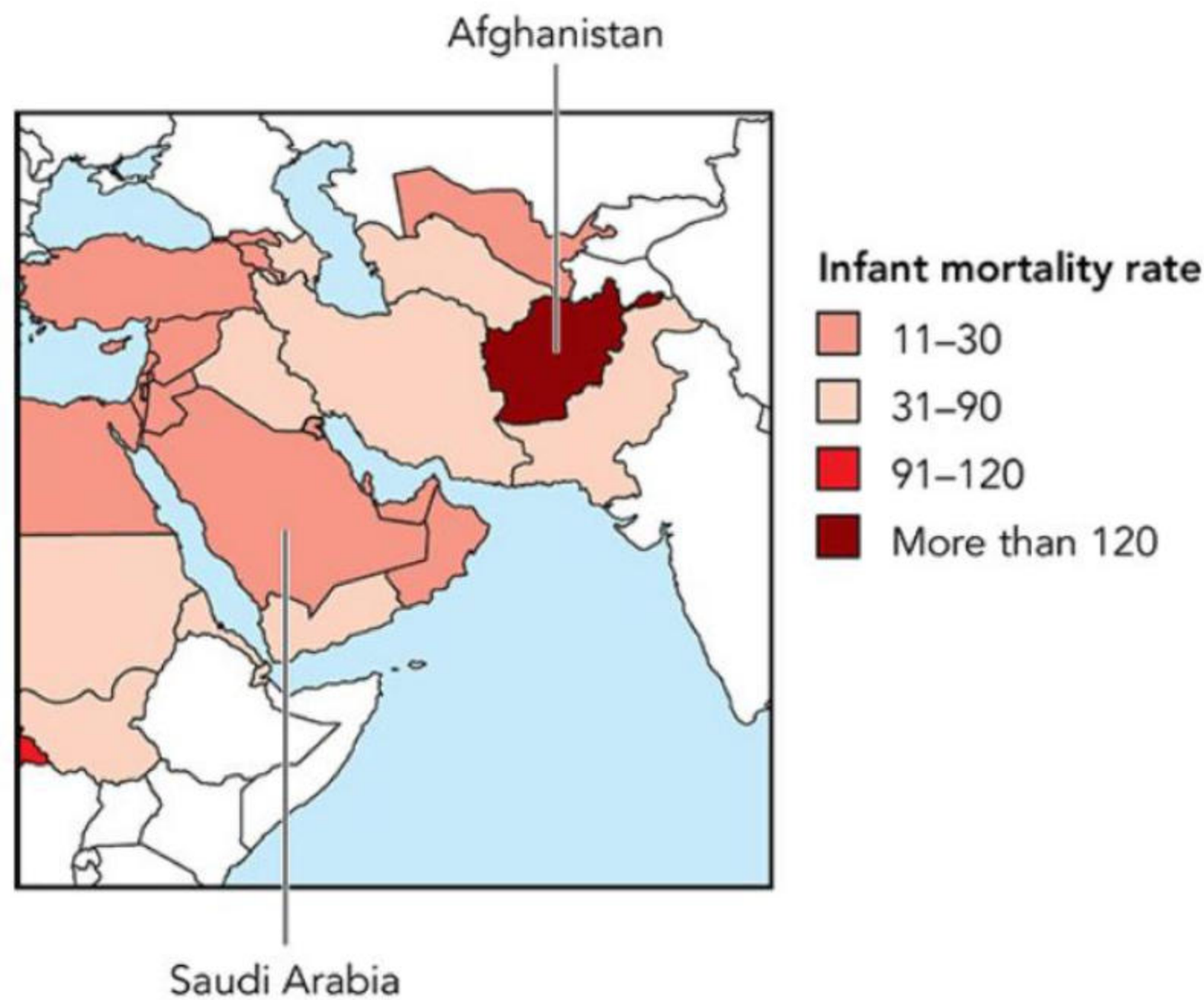


Figure 3.1

- i What is meant by infant mortality rate? [1]
- ii Using information from Figure 3.1, compare the infant mortality rate of Saudi Arabia and Afghanistan. [2]
- iii Suggest reasons for the differences in the infant mortality rate of Saudi Arabia (a HIC) and Afghanistan (a LIC). [3]
- iv State two other indicators of development. For each one, explain how it can be used to assess the level of social development of a country. [4]

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- b Study Figure 3.2, which shows a range of strategies to address differences in levels of economic development.

Top-down development strategies	Bottom-up development strategies
International targets, for example, United Nations Sustainable Development Goals	Work of non-governmental organisations
Aid	Microfinance
Trade	
Transnational Corporation investment	

Figure 3.2

- i Choose one bottom-up development strategy. Describe how it can help close the development gap between countries. [3]
 - ii Evaluate the advantages and disadvantages of top-down development strategies. [5]
- c For a named country you have studied, describe and explain the reasons for its current level of economic development. [7]

Author-written question

> Topic 9 Economic change

4

- a Study Figure 4.1, a map showing the island of Zakynthos, Greece (a HIC in Europe), along with Figures 4.2, 4.3, and 4.4, photographs taken in part of Zakynthos where the tourist industry is important.



Figure 4.1



Figure 4.2



Figure 4.3



Figure 4.4

i Which **one** of the following statements is correct about the distribution of tourist resorts in Zakynthos? Tick () your answer in the table:

	tick (✓)
--	----------

They are evenly distributed around the coast. They are in all parts of the island.	
They are inland in the north and west.	
They are on the coast, mainly in the east and south.	

- ii

What is the distance, in km, and direction **from** the airport **to** Alykes?

[1]
- iii

Using Figures 4.2, 4.3, and 4.4, identify **three** different ways in which Zakynthos benefits **economically** from tourism.

[2]
- iv

Suggest how the **infrastructure** of Zakynthos has been developed as a result of tourism.

[3]
- b

Study Figure 4.5, a graph showing information about the climate of Zakynthos.

[4]

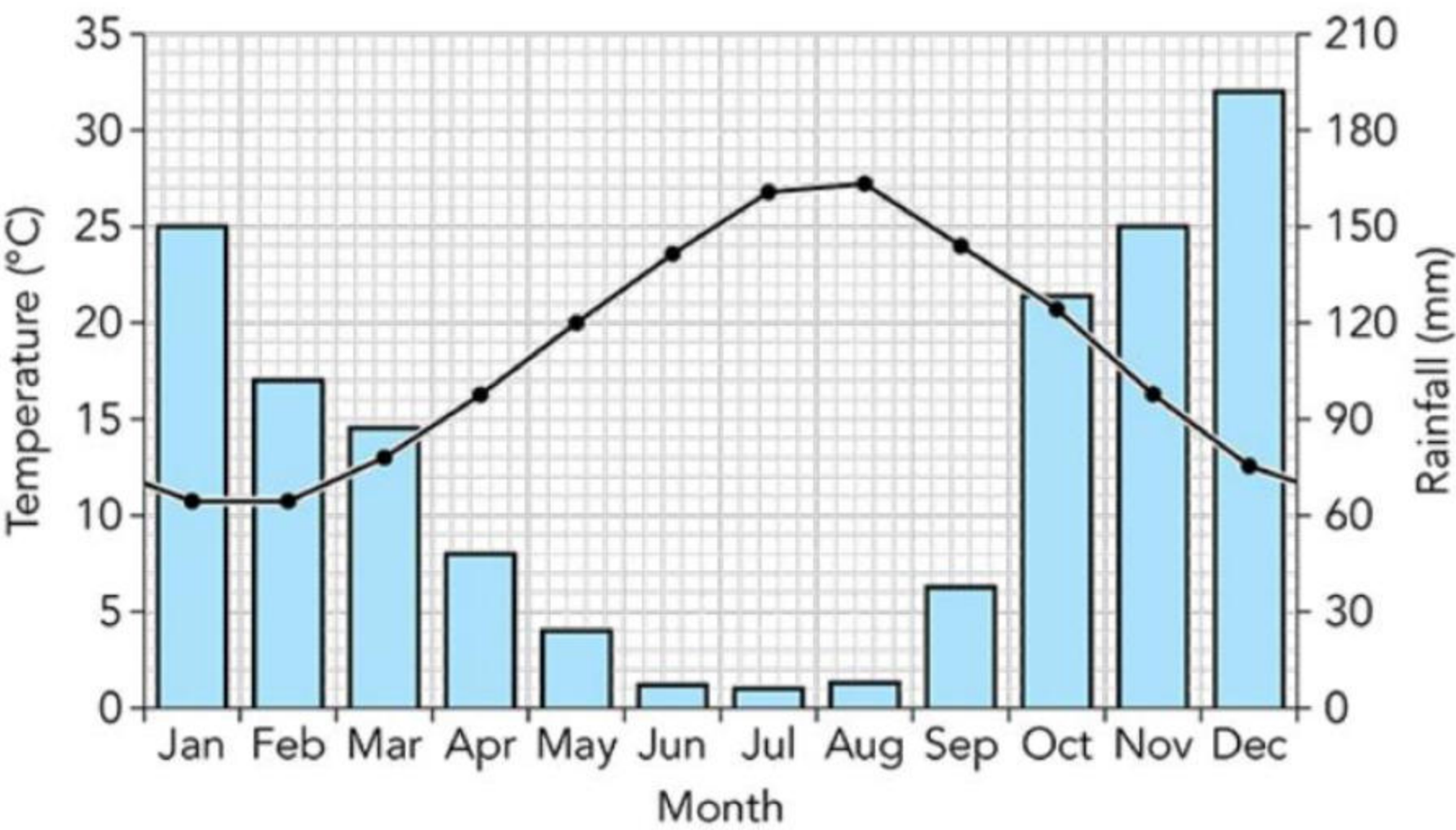


Figure 4.5

- i

Suggest reasons why the main tourist season in Zakynthos is from May to September. Use statistics to support your answer.

[3]
- ii

Every year 900,000 tourists travel to Zakynthos. 95% of tourists arrive between May and September. Suggest what problems there may be for people who live on Zakynthos, caused by large numbers of tourists arriving between May and September.

[5]
- c

For a named area you have studied where tourism is important, describe the attractions of the physical and human landscape.

[7]

> Topic 10 Resource provision

5

- a Study Figure 5.1, showing the different sources of energy for fulfilling energy demand in a country.

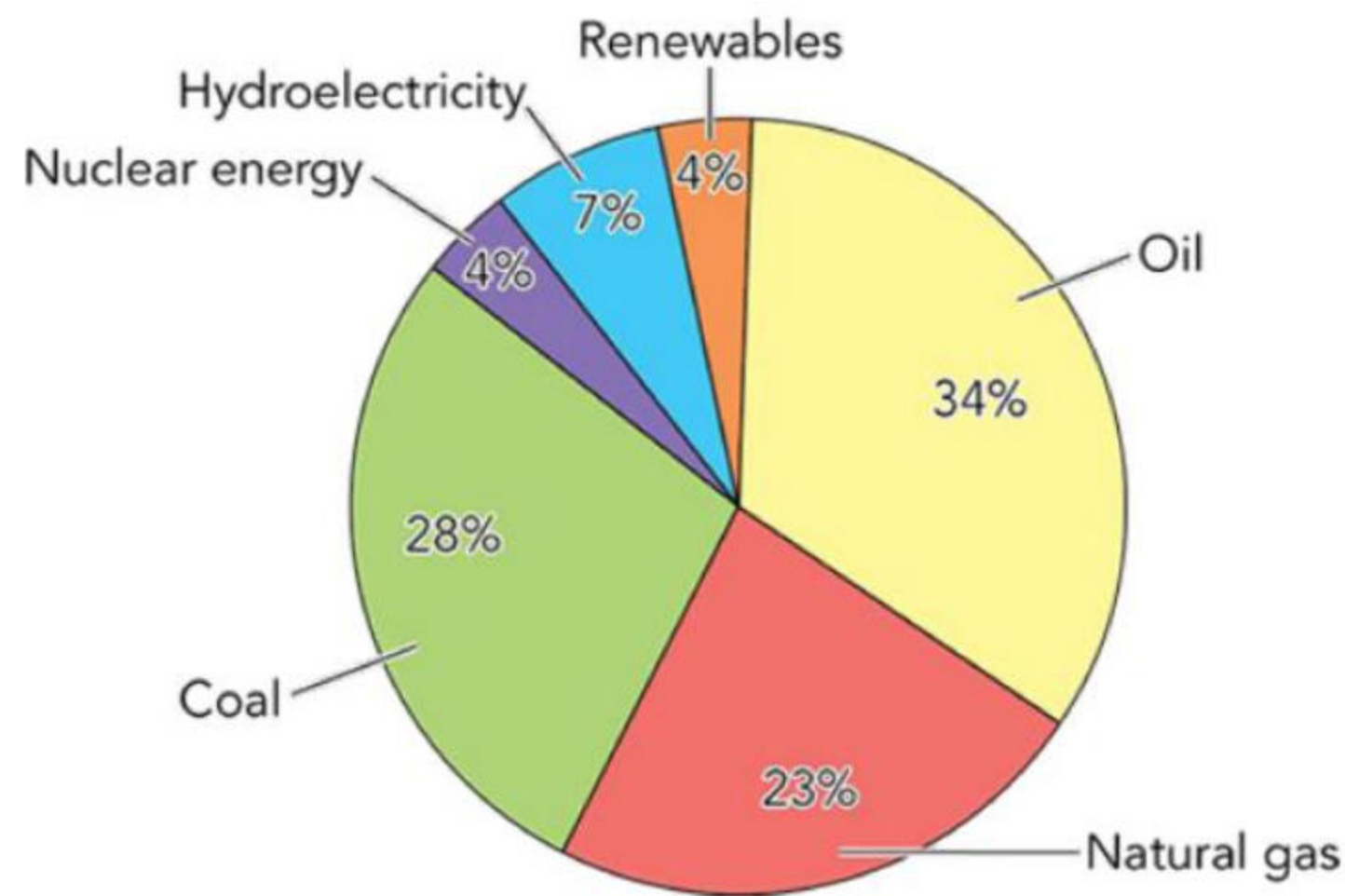


Figure 5.1

- Calculate the total percentage of energy generated from non-renewable energy sources. [1]
 - Give two examples of a renewable energy resource. [2]
 - What is meant by the term 'fossil fuels'? [1]
 - Suggest reasons for some countries generating more electricity from renewable resources than other countries. [3]
 - Describe strategies that a country can use to improve energy security. [5]
- b Study Figure 5.2, showing the top ten sweet potato producing countries in 2019.

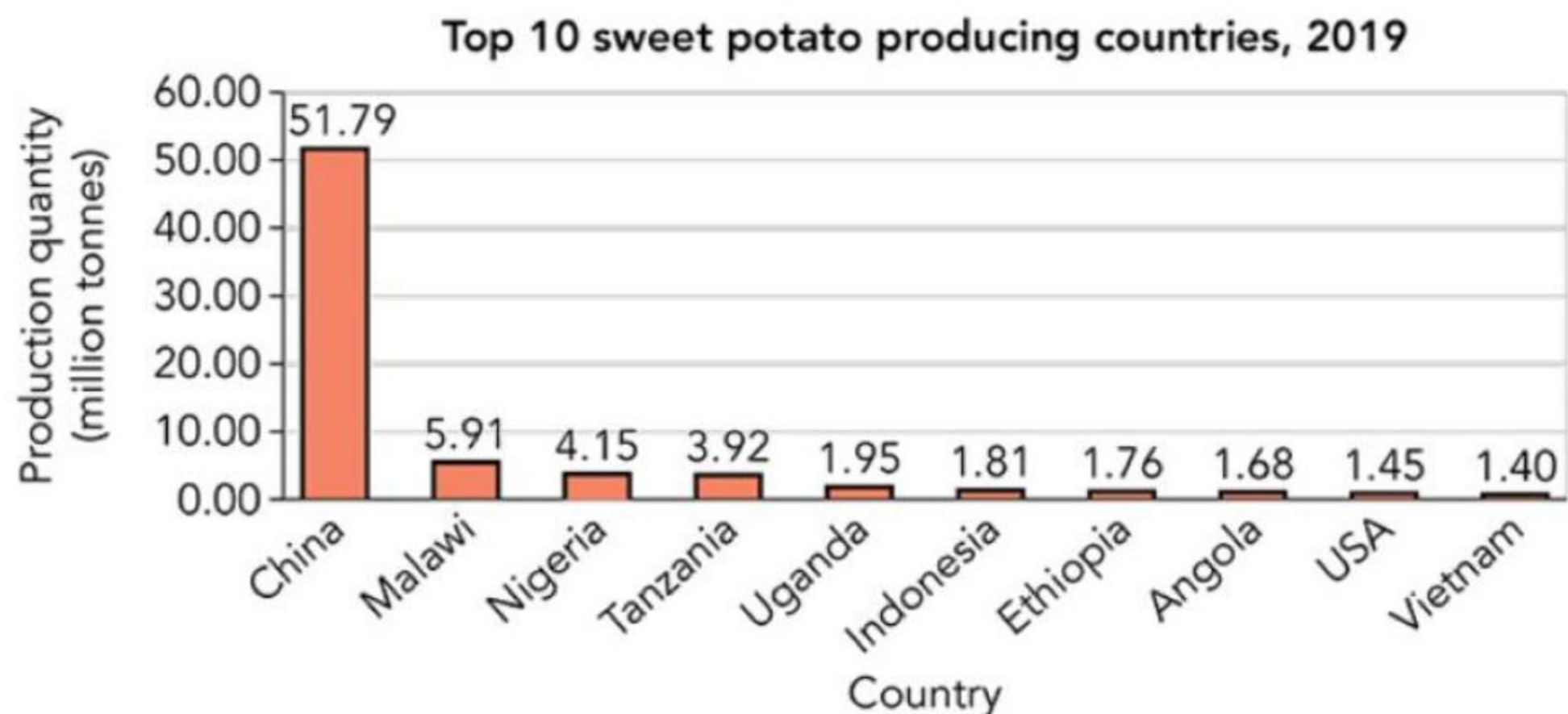


Figure 5.2

► Image Description

- Calculate the difference in sweet potato production between Malawi and Nigeria. [1]
 - A large percentage of the sweet potatoes produced in LICs are grown by subsistence farmers. What is meant by 'subsistence farming'? [1]
 - Suggest and explain two methods that farmers could use to increase their sweet potato yield. [4]
- c For a named country or region that you have studied, explain the impacts of food shortages. [7]

Author-written question

Geographical enquiry

Geographical enquiry

Geographical enquiry

LEARNING INTENTIONS

By the end of this unit you will be able to:

- understand the steps in planning fieldwork using the scientific method
- formulate and evaluate aims and hypotheses
- choose appropriate data collection methods
- consider your safety and required equipment for fieldwork
- make conclusions from the data you collect.

GETTING STARTED

In pairs, consider what you would need if you were going on a field trip. Note down:

- the steps you would take before a field trip
- the steps you would take during the field trip
- the steps you would take after a field trip.

When studying geography, you are learning about people and places. You need to be able to carry out investigations about these issues of human and physical geography. These investigations will include collecting information about questions you or someone else has asked. For this course, you must understand how to carry out human and physical geography investigations in a structured and scientific manner.

There are two pathways for assessing your fieldwork skills in this syllabus:

Coursework

You can base your coursework on either physical or human geography, or a combination of the two. The aim of your coursework assignment is to test your knowledge and understanding of geographical topics, your geographical skills and data analysis, and your judgement and decision-making skills.

Geographical Investigations

If you don't have time for data collection or there are a lack of suitable fieldwork sites, you can take the Geographical Investigations paper. The emphasis of this paper is on testing your geographical skills and enquiry skills, and your ability to analyse information. The paper includes tasks that are related to the topics covered in the course and can include physical and human geography related questions.

For either option, you need to have a sound understanding of fieldwork and the scientific method, and the skills required to complete specific fieldwork investigations. This unit will outline the scientific method and examples of popular fieldwork investigations. Work through both the skills unit and this enquiry unit to give you guidance for your methods and how you want to analyse and present your findings.

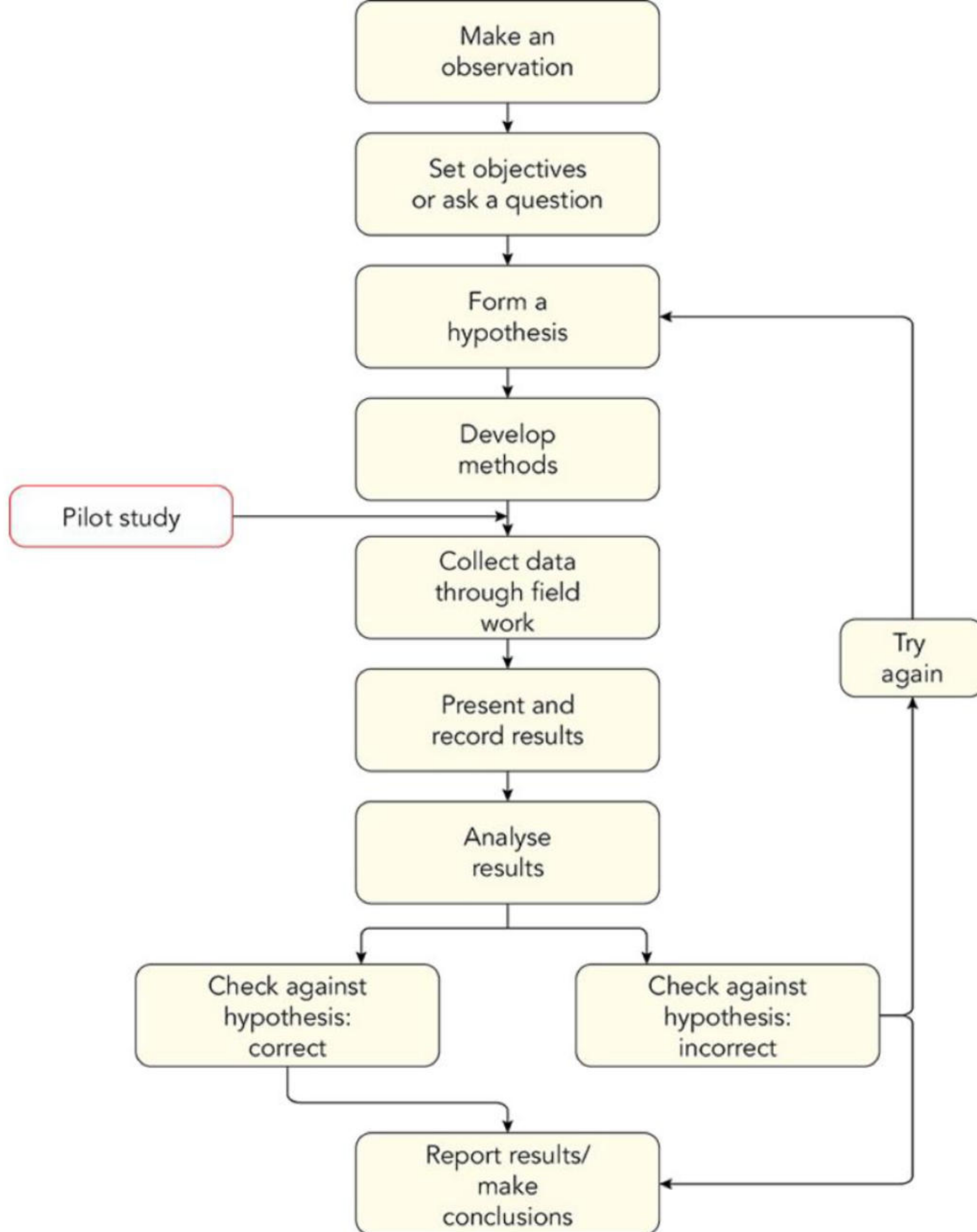
Fieldwork and the scientific method

When completing fieldwork, there are a number of steps you need to take (Table 11.1 and Source A). These steps are part of the scientific method. Following these steps ensures that you are collecting reliable, unbiased data that can be used to compare change at a site overtime, or information between sites. This unit takes you through these steps, as well as example topics for fieldwork. Presenting, recording, and analysing results is covered in the 'Introduction to geographical skills' unit.

Table 11.1: Steps in planning fieldwork using the scientific method for geographical investigations

Step	Description
Make an observation	Identify what you are investigating. For example: 1 The movement of the population in a city. 2 A river profile.
Set an aim/objective, or	Write a question or aim related to the observation you made. For example:

ask a question	1 Does the number of pedestrians using the areas between 12pm and 1pm vary each day of the week? 2 How does the river profile change downstream? The question or aim should be simple. It should identify one question that you want to find an answer to. If you ask too many questions, you may find that you do not clearly answer any of them, and your data may become difficult to analyse.
Form a hypothesis	Create a statement that predicts what you think you will find. For example: 1 There are more pedestrians on a Saturday than on the other days of the week. 2 The river becomes shallower and wider downstream.
Develop methods	Develop the methods you are going to use to ensure that you are able to answer your hypothesis.
Collect data through fieldwork	Collect data to test your hypothesis. The data collection methods will vary depending on what you are investigating.
Present and record results	Use information from the skills chapter to review how to present data. For example, you can present results in tables or graphs.
Analyse results	Use the information from the skills chapter to help you analyse the results, using statistical analysis and comparison of information, to see what the data tells you.
Check results against hypothesis	Check if the data supports your hypothesis. The information that you collect will prove, partially prove, or disprove your hypothesis. If your data does not support your hypothesis, you start again, resetting a hypothesis and retesting.
Report results/make conclusions	Report your findings, whether the hypothesis is true or false. Others will then be able to refer to this information to guide them in the future.



Source **A**: The scientific method in geography fieldwork

Forming and evaluating a hypothesis

A hypothesis is a statement of an expected result when collecting data to analyse an observation. A hypothesis can be proven true, partially true, or false. It is a possible answer to a 'how' or 'why' question.

A good hypothesis is a simple statement, not a question. To write a hypothesis, start with a question, for example, 'What do farmers in the district of Trent grow?'. Then form a testable statement from the question, for example, 'Most farmers grow bananas in the farming district of Trent'.

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To evaluate a hypothesis, you need to have completed the collection of information, you then use the data to determine if the hypothesis was true, partially true, or false. For example, for the hypothesis 'Most farmers grow bananas in the farming district of Trent', you collect the following data: 20% of the farmers grow bananas, 52% grow coffee, and 28% are pastoral farmers. The hypothesis would be proven false. When it is found to be false, do not alter the hypothesis to fit the data. Instead, use that data to show how or why the hypothesis was false.

TASK 1: Study Sources **A** **B**

a In pairs, create a hypothesis for the following observations. Think about your reasoning for why you are setting the hypothesis:

i Are the prices of goods more expensive in large grocery stores or in small convenience stores?

- ii Do people use the cycle lanes that have been provided in a town?
 - iii Is the slope that has been undercut by a road stable?
 - iv Does pollution of the River Garrick increase downstream from a factory?
 - v Compare your hypotheses with another pair and discuss your reasoning for your hypotheses.
- b With reference to Source A, explain the steps you would take if you found that the data proves a hypothesis to be false.
- c With reference to Source B, state whether the hypothesis ‘Shop 2 is larger and busier than Shop 1’ is true, partially true, or false.

Learners investigated the shopping habits of shoppers in two shops in the town of Logan.

The hypothesis for the investigation is: ‘**Shop 2 is larger and busier than Shop 1**’. Learners visited both shops and collected data using the following techniques:

- interviewing the managers
- counting customers
- measuring the fronts of the shops
- studying the car parking areas.

The data collected is show in the table below:

Data collection method	Shop 1	Shop 2
Site location an interview with manager	¼ km from the CBD	2 ¼ km from the CBD
	Located on a busy road	Located at a large intersection of four main roads
	Employs 5 staff	Employs 241 staff
	Privately owned	Owned by transnational corporation
Shop observations from site visit	Sells mostly convenience items	Sells convenience and comparison items
Measurement of shop frontage	10 meters	45 meters
Customer count for 15 minutes at midday	52 people	421 people
Car park spaces	8 cars	300 cars
Number of cars outside during visit	7 cars	219 cars

Source B: An example of a geography fieldwork investigation

Investigator bias

Investigator bias is when a scientist knowingly or unknowingly selects data that encourages one outcome or answer over another. This bias can occur at any phase in research, including the design of the study, collection of data, and the processing of the data. For example, a student carrying out a survey on most popular mobile phone brands is nervous about asking adults questions, so they only approach other students. This is an unintentional bias that will influence the data collected and the outcomes in the study; they will find out the most popular mobile phone brands for students, not the general population.

As a scientist you need to be able to look at data of other researchers and determine if there is bias in their research and how it influences conclusions.

Collecting data

There are different factors to consider when collecting data in the field. These include the safety of the team and data collection methods.

Fieldwork safety

Taking part in fieldwork poses risks and hazards that you need to identify before the start of the investigation. When you identify risks correctly, you reduce the chance of accident or injury.

Things to consider when reviewing the risk of a field trip:

Ensure clear and effective communication

- share emergency contact details

- always have a means of communicating with you at all times
- work in pairs or groups.

Consider environmental conditions

- dress appropriately
- bring sunblock and insect repellent
- be cautious around water, wild life, busy roads, and cliffs
- avoid unsafe or dangerous areas.

Prepare for medical issues

- share any relevant medical concerns with your teacher and group
- bring a basic first aid kit
- pack enough food and water.

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Source **C**: A market street showing shops and road traffic



Source **D**: A collapsed coastal road along a sea cliff in Skipsea, UK

TASK 2: Study Sources **B** **C** **D**

- a** With reference to Sources B and C, list three pieces of advice that the teacher should give the learners before they start their fieldwork.

b With reference to Source D, suggest three safety precautions that are different to those you identified for the investigation taking place in part a of this task.

Fieldwork checklist

Fieldwork checklists help you to ensure that you are prepared for a field trip before departure. In many instances, fieldwork can be in remote locations. Returning to collect a forgotten item is not always possible! You need to consider the following:

- a clearly identified aim and hypothesis/hypotheses
- what **primary data** you will be collecting
- the **secondary data** you have access to
- what type of sampling you will use (**systematic** or **random**)
- how you will analyse and present data
- methods you will use to test the hypothesis
- identification of all the equipment you will need for the investigation, this is based on the methods that are identified for the investigation
- safety checks (health, safety, and welfare).

Table 11.2: Different items that may be required for a field trip

Measuring equipment	• Quadrats: Placing a square of a known size (the quadrat) either randomly or at regular intervals within a study area, in order to count and identify the species density, or diversity in a given location. • Transects: Setting up a straight line across a habitat or sample site to collect data about the distribution and abundance of a species. • Floats: any material that floats that is used when measuring the speed of flow of water. • Noise meter: Equipment that is used to determine the intensity of sound in an area. • Stopwatches: Equipment used to measure the time of an event or experiment. • Digital instruments: Instruments that measure, analyse and record data in the digital form.	• Clinometers: An instrument used for measuring the angles of a slope. • Callipers: a measuring instrument with two arms that can be adjusted to determine the distance between two points, or the size of an object. • Rulers: A tool for measuring the straight line length of an object • Measuring tapes: a long tape marked with measurements, used to set or determine distance. • Ranging poles: A straight rod or pole, painted in bands of alternative colours of red or white and used for measuring slope. • Click counters: A tool used for counting the number of something by keeping a total of clicks of any object being counted.
Observations	• field sketches • photographs	• recording sheets
Questionnaires (things to consider)	• the layout • the number of questions • how large the sample size is	• the format of questions • the target audience • where it will be carried out • what problems may be faced
Secondary data	• books • libraries • internet research	• newspapers • census data • previous research
Presenting data	• the type of table used • the statistics used	• the types of graphs used • the report layout

When carrying out human geography investigations, we often need to collect information from members of the public for their opinions and views through the use of surveys. A survey is a method of research that is used to collect data from groups or individuals to gather information on experiences, preferences, opinions and population demographics. They can be conducted in various formats which include questionnaires, forms, interviews, bipolar surveys and environmental quality surveys.

When designing surveys it is best to test that your questions/survey forms are appropriate prior to carrying out the survey as a **pilot study**, as this can save you time and money in the long-term

More specific data collection methods can be found under the 'Example topics for coursework' section.

Questionnaires and interviews for data collection

Using a questionnaire or interview can be a good way to gather this type of information as it gives insight into people's preferences and habits. Planning is important here, as it ensures that you make sure your questions collect information that is relevant to your study.

Questionnaires

A questionnaire is an investigative tool in which a series of questions are set for the target market to answer. There are three main types of questions:

- 1 multiple-choice questions (respondents have a choice of answers)
- 2 closed questions (yes or no answers)
- 3 open questions (questions that seek a longer answer or opinion from the respondent).

Limitations of a questionnaire are that some questions may be unanswered or dishonestly answered. The person answering the questions also might not understand the question or may interpret the questions differently. However, questionnaires are inexpensive, practical, and can be easily used to get fast results.

In order to get the best results from a questionnaire, they are best done face to face. However, they can also be sent online or done over the phone. When carrying out questionnaires, consider the following:

- don't inconvenience respondents:
 - keep questionnaires short, simple, and keep personal information general
 - be polite
 - explain the purpose of your survey
- record the location and time of your survey
- be conscious of your safety
- consider a pilot study.

Interviews

An interview is the collection of data through direct conversation between the investigator and a respondent on a one-on-one basis. This style of questioning allows you to obtain more in-depth information. An interview is made up of open questions. This is a more expensive method of obtaining data than a questionnaire, as it is more time consuming to collect and analyse. The data is also more limited. The advantage is that with face-to-face interviews, the answers are often more accurate.

Mall questionnaire

Investigator's name:

Location:

Date:

Time:

Please tick the relevant box for your answer.

1) What is your age bracket?

18-24 ☐ 25-34 ☐ 35-44 ☐ 45-54 ☐ 55-64 ☐ 65 and over ☐

2) How often do you shop at the mall?

Once a month ☐ Once a week ☐ More than once a week ☐

3) What are the main reasons for visiting the mall?

Grocery shopping ☐ Clothes shopping ☐ Restaurants ☐

Entertainment ☐ Meeting friends ☐

4) How do you feel about the prices at the mall?

Reasonable ☐ Not reasonable ☐

5) Would you recommend the mall to others?

Yes ☐ No ☐

6) What is your favourite store at the mall?

7) Why is that your favourite store at the mall?

Source E: A questionnaire designed by the owner of a mall to collect information about shopping and experience preferences

TASK 3: Study Source E

- a Identify one open question and one closed question.
- b Create a closed question that finds out more about the distance people travel to visit the mall.
- c Explain why the distance that shoppers travel would be important information for the mall owner.
- d In pairs, write three interview questions that you could ask the mall manager that would reveal more about the shopping preferences of the shoppers.

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Land use surveys

Land-use surveys collect data about what a particular piece of land is used for, for example, the number and types of buildings in a particular area, or how land use changes as you move further away from the CBD. This information can be added to a database or map to understand how a particular place is being used.

Bipolar surveys

A bipolar survey is a type of questionnaire designed to assess the opinions of people where there are opposing views. This survey allows the responders to express their views regarding a statement or topic, along a scale of two extremes.

For example: How do you feel about the options on the menu?

1	2	3	4	5
Very dissatisfied	Dissatisfied	Neutral	Satisfied	Very satisfied

Responders can select the number that best expresses their viewpoint.

The benefits of this type of survey is that it is quantitative and qualitative, making for easier interpretation of results.

Environmental quality surveys

An environmental quality survey is an assessment of a variety of environmental factors within a specific geographical area. The aim of such a survey is to gather data about the biological, physical and human built environments. This type of survey allows investigators to understand the current standard of a given environment and helps to identify areas that require improvement.

Table 11.3: An Environmental survey of a town, investigating the quality, cleanliness and safety of the area using a ranking scale that goes from a negative evaluation to a positive evaluation.

Negative evaluation	-3	-2	-1	+1	+2	+3	Positive evaluation
Green areas absent							Green areas present and well maintained
Roads and pavements damaged/absent							Roads and pavements well maintained
High noise levels							Low noise levels
High levels of air pollution							Low levels of air pollution
Heavily littered with rubbish							No obvious litter present
Abundant graffiti							No evidence of vandalism
Very little space and buildings close together							Open spaces with buildings further apart
Low quality buildings							High quality buildings
Unsafe							Safe
Threatening							Welcoming
Total Score							

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The students study the area and complete the survey, and then tally up the score given, the lower the score the lower the environmental quality. As this is a subjective survey it is beneficial to have more than one team carrying out the survey then comparing results or working out averages.

TASK 4

As a class create a survey for your school, or the surrounding areas. Then carry out the survey in teams of three or four. Once you have completed the survey compare your findings. Were they similar or very different? Discuss the outcomes to understand what the cause of differences in the survey may be.

Drawing conclusions

A conclusion is when you make a reasoned judgement using the data that you collected during field work. To form a conclusion, take the following steps:

- 1 Revisit the hypothesis to determine what you were testing.
- 2 Write a clear statement about the data that supports how strongly the hypothesis was found to be either true or false.
- 3 Use data clearly to support your argument.
- 4 Acknowledge any unusual results and if possible, explain the reason for them.

TASK 5: Study Source B

Write a conclusion to the investigation outlined in Source B.

Remember to use data to support your findings for the hypothesis that ‘**Shop 1 is larger, busier and more important than Shop 2**’. Clearly state if you find the hypothesis true, partially true, or false in your conclusion.

Example topics for coursework

The rest of this unit focuses on four popular topics for fieldwork investigations. It includes guidance for data collection specific to the topic and example enquiry tasks.

Investigating the CBD

Towns and cities give geographers a number of opportunities for fieldwork. There are different factors you can investigate in a town or city’s Central Business District (CBD). For example, you can investigate:

- where the CBD starts and ends
- if and how the function of buildings changes with height or in different parts of the CBD
- how well pavements allow for the flow of pedestrians
- the level of congestion on the roads
- population density during different times of day.

Source F shows an example enquiry for a fieldwork investigation. You could use it as the basis for a fieldwork task in your local setting.



Figure 11.1: Phoenix, Arizona, USA, with the CBD visible at the centre of the city

The Netherlands is a relatively flat country and cycling is a very popular mode of transport. Amsterdam has an estimated 800,000 cyclists (63% of the population) that use their bicycles daily. 32% of the traffic movement is by bicycle, compared to only 22% by car and 16% by public transport. In the CBD, 48% of the traffic movement is by bicycle.

Amsterdam has 767 km of cycling paths and lanes, and it is estimated that 2 million km is cycled daily by cyclists. Cycling is seen as an answer to the speed at which commuters can move around the CBD while also being a low-carbon commuting option that reduces air pollution in and around the city.



Figure 11.2: Cyclists ride their bicycles in Amsterdam

The enquiry:

Enquiry question: how many bicycles with specific categories pass by a selected point on a cycling path?

Equipment: high visibility vests, clipboard, watch, recording sheet, pens, camera, 16 learners.

Method: between 5pm and 5:20pm (20 minutes), eight pairs of learners will carry out a bicycle traffic count. They will use the following categories:

- 1 bicycles with adult passengers
- 2 bicycles with child passengers
- 3 children on bicycles (no passengers)
- 4 adults on bicycles (no passengers).

Pairs of learners will stand on opposite sides of the cycle path to count. They will use the tally method on a recording sheet.

Safety: high visibility clothing, mobile phone, teacher, assembly point, working in pairs, select safe position.

TASK 6: Study Source F

- a Read the plan in Source F. Do you think it could be improved? Consider the categories the learners are identifying, and the potential for the use of technology for counting.
- b Write a hypothesis for the following question: which category of cyclist is likely to be the highest?
- c The students collected the following numbers in the 20 minutes that they stood at the location:
 - bicycles with adult passengers: 29
 - bicycles with child passengers: 41
 - children on bicycles (no passengers): 35
 - adults on bicycles (no passengers): 285.

Use this data to create a graph to represent their findings. Decide on the type of graph and the scale that you need to use.
- d Was your hypothesis correct or incorrect? Explain why you think this data could have been anticipated.
- e In pairs, design another investigation to find out information about how the number of pedestrians changes at different times of the day in your town or city.

THINK LIKE A GEOGRAPHER

Geographers collect data and work with city planners to consider the rate of growth of a city, and how that expansion affects the land within and surrounding the city. They need to use this information to consider how we can manage cities for future development. In pairs, discuss what you think the city planners need to consider when they are structuring urban management, for example, traffic flow.

Investigating tourism

Understanding tourism and the movement of people is necessary for businesses, towns, and countries to meet the needs of tourists and retain them as customers. Tourism can be a good focus for an enquiry. There are many sets of data that you can analyse to investigate how many people are visiting a region. For example, you can track the number of travellers arriving by air, sea, or road, or analyse how full hotels are at different times of the year.

Example enquiry

Madagascar is an island nation off the east coast of southern Africa. It relies on income from eco-tourism to help support its economy. Income from tourism amounted to only 2.8% (\$106 million) of the GDP in 1995 but had increased to over 6% by 2019 (\$951 million). The COVID-19 years saw this decline to below 1% (\$111 million) by 2021. The tourist attractions to Madagascar include the beaches and wildlife. The historical sites and relaxed cities with products from local craftsmen result in it being a return destination for many travellers.

Source G shows the climate data for Madagascar, giving an overview of the temperature and rainfall in Madagascar throughout the year. Climate data is an example of the type of information you could collect to gain fieldwork experience in your local setting.

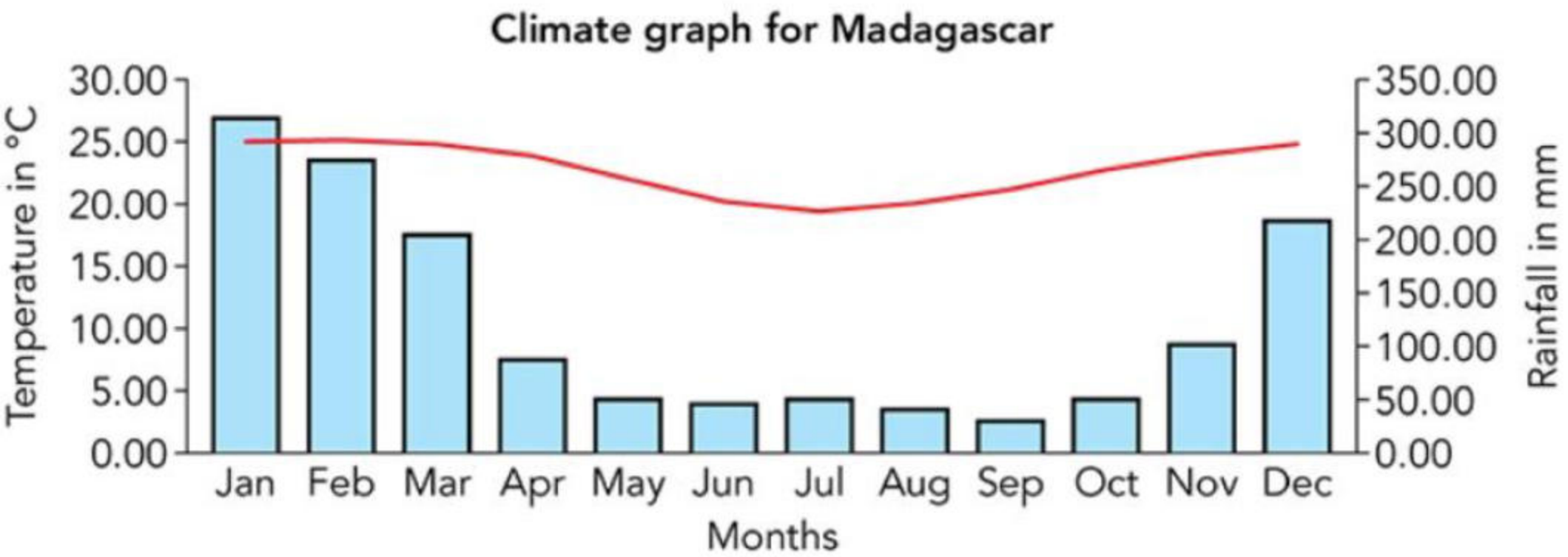




Figure 11.3: The ring-tailed lemur is one of approximately 110 species of lemur which are endemic (found only on that island) to the island of Madagascar

TASK 7: Study Source **G**

- a** With reference to Source G, suggest the best time of year to visit Madagascar. Use data to support your argument.
- b** Considering tourism, do you think the Madagascan government should invest in protecting their natural ecosystems? Explain your answer.
- c** In pairs, create a questionnaire to find out more about the type of tourist that visits Madagascar. Consider why they travel there, where they travel from, how long they visit for, and who they travel with when planning the questions. Test your questionnaire on other learners in the class or school. Refer back to Source E on questionnaires as a guide.

Investigating rivers

River investigations can focus on one location on a river. They can also compare information along the length of the river, which involves taking samples from different points along the river. Examples of river features you can measure include:

- width and depth
- river velocity
- river load
- changes in gradient.

Measuring river features and processes is fun but can be quite dangerous. Much of the fieldwork involves working within or near a river. This means you need to find areas with easy access to shallow water in which to work.



Figure 11.4: A student taking water samples, while wearing waterproof waders to stay dry

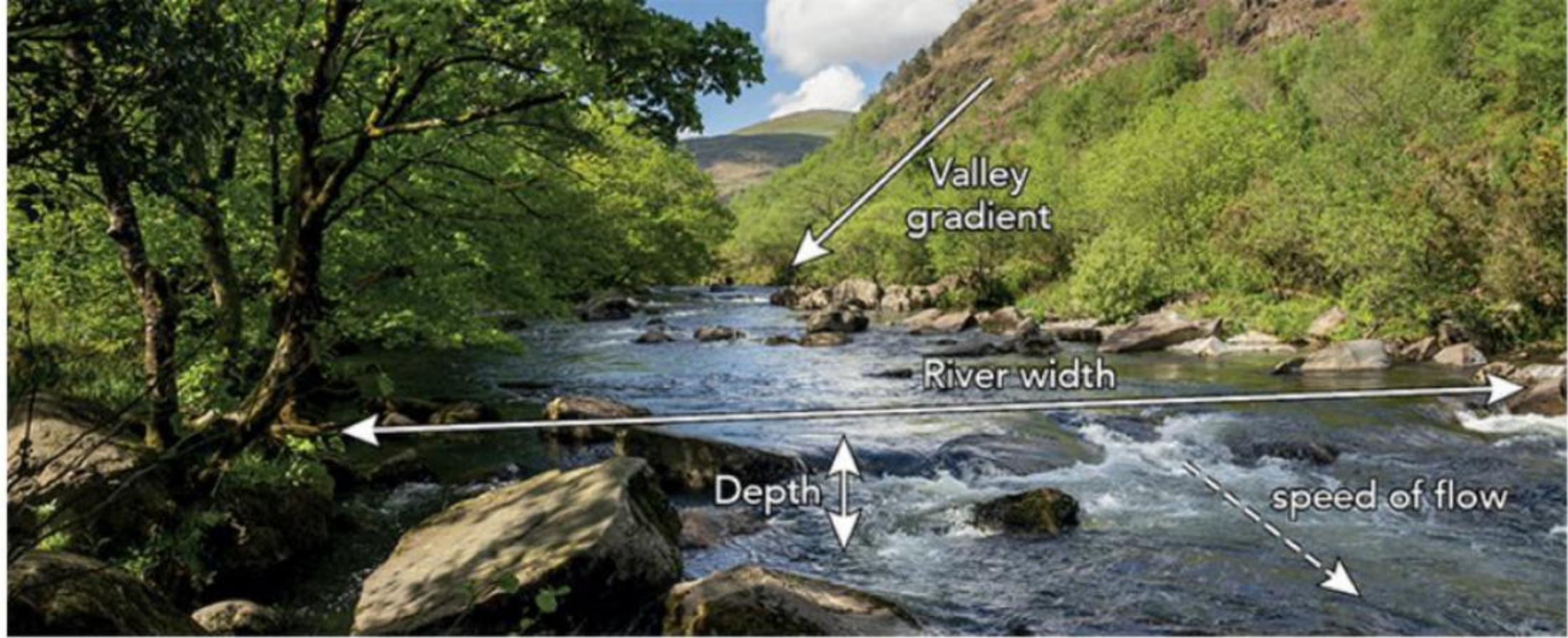
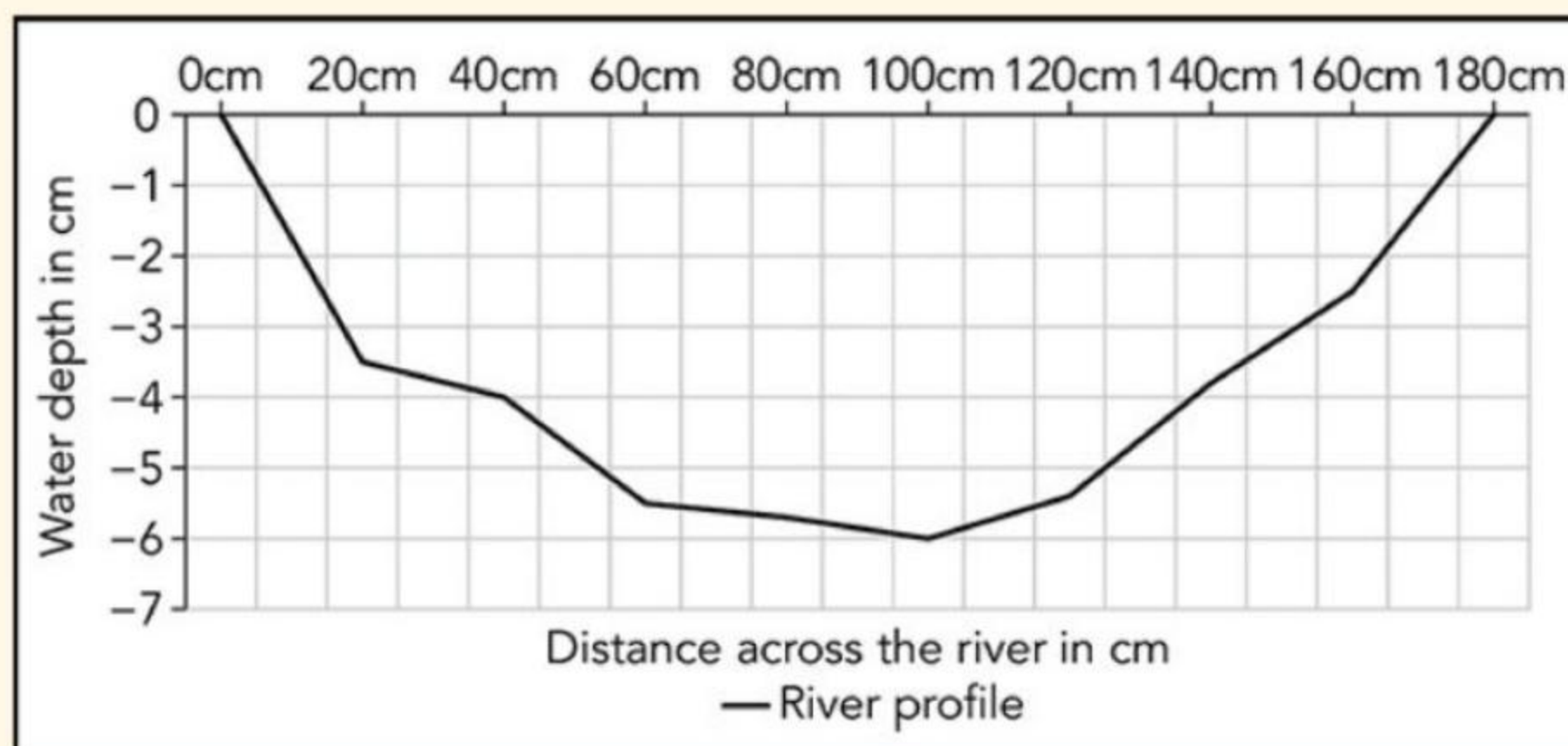


Figure 11.5: Some of the different factors one can measure during river fieldwork

Measuring width and depth

To measure the width and depth of a river, you use a tape measure and a metre ruler:

- 1 Measure the width from the edge of the water on one side to the edge of the water on the other side. This is easiest to do in pairs, with the tape measure stretched tautly across the river to get the river width.
- 2 Measure the depth of the river at a number of points across the width of the river. The distance between measuring points for small streams and rivers can be every 20 or 25 cm.
 - a Stretch the tape measure across the river. You can secure it in place with two people holding it, or with pegs.
 - b Take a reading of the depth of the river every 20–25 cm, recording the depth each time (Source H).
 - c Use this information to plot a cross section of the river (Source H). Plot every measurement that is below the water level as a negative value, as we consider the top of the water as 0 cm.



Source H: An example of a river profile

Distance across the river	Depth measured (cm)
0 cm	0
20 cm	-3.5
40 cm	-4
60 cm	-5.5
80 cm	-5.7
100 cm	-6
120 cm	-5.4
140 cm	-3.8
160 cm	-2.5
180 cm	0

Measuring river velocity

To measure the velocity of a river, you can use a flow meter. A flow meter has a propeller that turns in the water, giving a reading of the speed in metres per second. If you don't have a flow meter, you can use a stick/a float/some dye, as well as a tape measure and stopwatch. Then take the following steps:

- Measure and mark out a distance of ten metres.
- Throw the object in above the start of your measured ten metres. When it passes the start point, start timing using your stopwatch.
- Stop timing when the float has travelled the ten metres.
- Repeat this five times and take an average to reduce the risk of errors.
- Calculate the velocity using the formula: $\text{Velocity} = \frac{\text{distance}}{\text{time}}$

Note that measuring velocity at various points along the river or at different depths across the river will give different flow speeds.

Measuring river load

The river load is bits of eroded material, generally rocks, that the river transports until it is deposited. As the velocity of the river decreases, the ability to carry the load decreases, and deposition occurs. One can therefore compare different sites, both the velocity and the load, to determine changes in the river.

You should take several random samples at several points across the river width at each sample site. Take the following steps:

- At each point, reach into the river. Point directly down with your index finger without looking to select a pebble.
- Pick up the first pebble or particle that your finger touches.
- If possible, measure the length of the pebble using callipers or a ruler and record the measurement.
- Repeat the process at least five times at each point, and at points across the river.
- Calculate the average size for the load at each sample point.
- If the riverbed is made up of mainly fine sediments, take samples in jars and let them settle. You can then make a visual comparison in how they differ.
- You can also estimate the roundness or smoothness of stones using the Power's scale of roundness (Figure 11.6).

If you are studying the river's long profile and need to measure the change in gradient as it flows downstream, measure the slope angle as described in the 'Investigating coasts' section below.

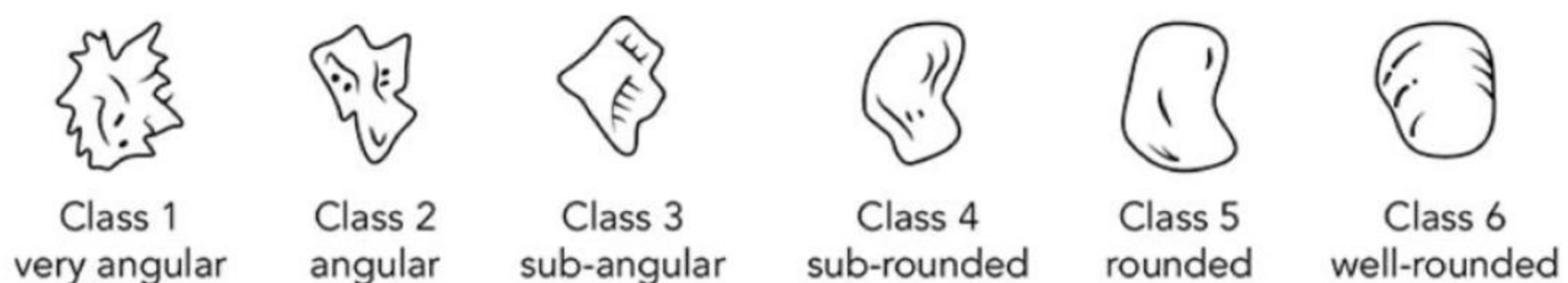


Figure 11.6: Power's scale of roundness

TASK 8

- To make measuring the river profile as easy and as safe as possible, how many people do you think you need? Think of all the steps you need to take to complete the data collection.
- Plot a river profile** for the data recorded in the table below.

Distance across the river	Depth measured (cm)
0 cm	0
20 cm	-1
40 cm	-2.5
60 cm	-7.5
80 cm	-10
100 cm	-12
120 cm	-11

140 cm	-7
160 cm	-3
180 cm	0

- c **Compare this river profile with the example in Source H.** Comment on both the shape of the profile and the difference in the depths of the river.
- d List all the items you need before going on a river field trip to measure width, depth, and velocity. Include any materials that you need to record data.
- e Create a hypothesis for what you think will happen to the size and shape of bedload the further you move from the source of the river.

Investigating coasts

Coastal investigations can focus on one location on a coast, or a long stretch of coast. They can also compare information along the coast, which involves collecting information from different sections of the coast. Examples of coastal features you can measure include:

- Beach profile
- Longshore drift
- Dune vegetation.

As with investigating rivers, there are risks when carrying out fieldwork in coastal regions. Take care to identify the risks and work carefully to avoid any accidents.

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Measuring a beach profile

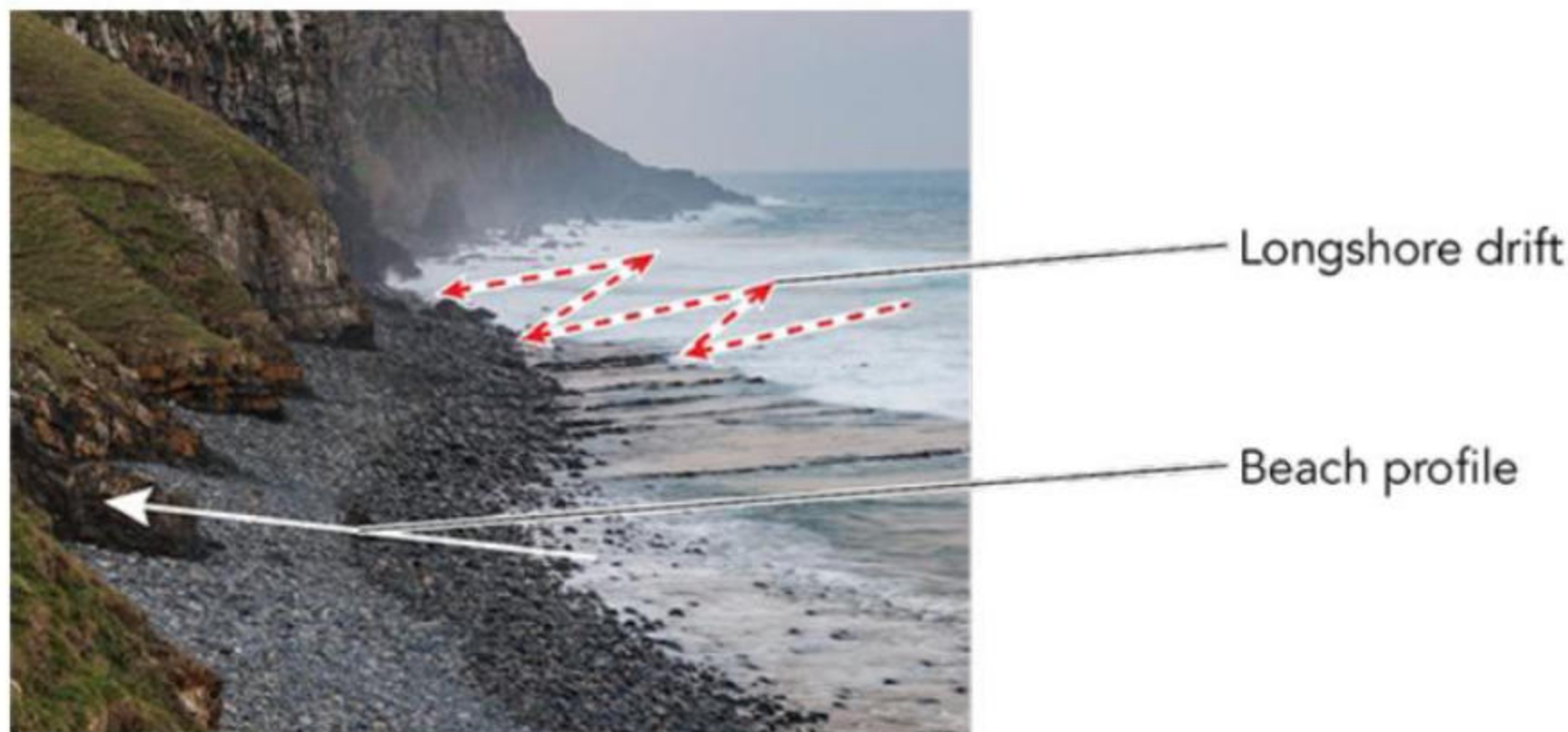


Figure 11.7: Some of the factors that one can measure when investigating coasts

For measuring a beach profile, you will need to select a number of sample sites along the beach. This will allow you to compare how the beach profile changes along its length. You will need ranging poles, a tape measure, and a clinometer. At each sampling site, take the following steps:

- 1 Work in pairs. One person should place the first ranging pole vertically at the high tide mark. The other person should place the second pole where the angle of the beach changes for the first time (Figure 11.8).
- 2 Use a tape measure to measure the distance between the ranging poles. Record the distance.
- 3 Use a clinometer to line up the top of equal bands on each ranging pole. Record the angle. Repeat this three times so that you can calculate an average and minimise the risk of error.
- 4 Repeat this step for every section of the beach where you note an angle change.
- 5 Once you have collected all the information, plot the data on graph paper to draw the beach profile. Remember to include the distance between each sample to ensure that your beach profile is drawn to scale.
- 6 You can then compare the beach profiles that you create for different parts of the beach.

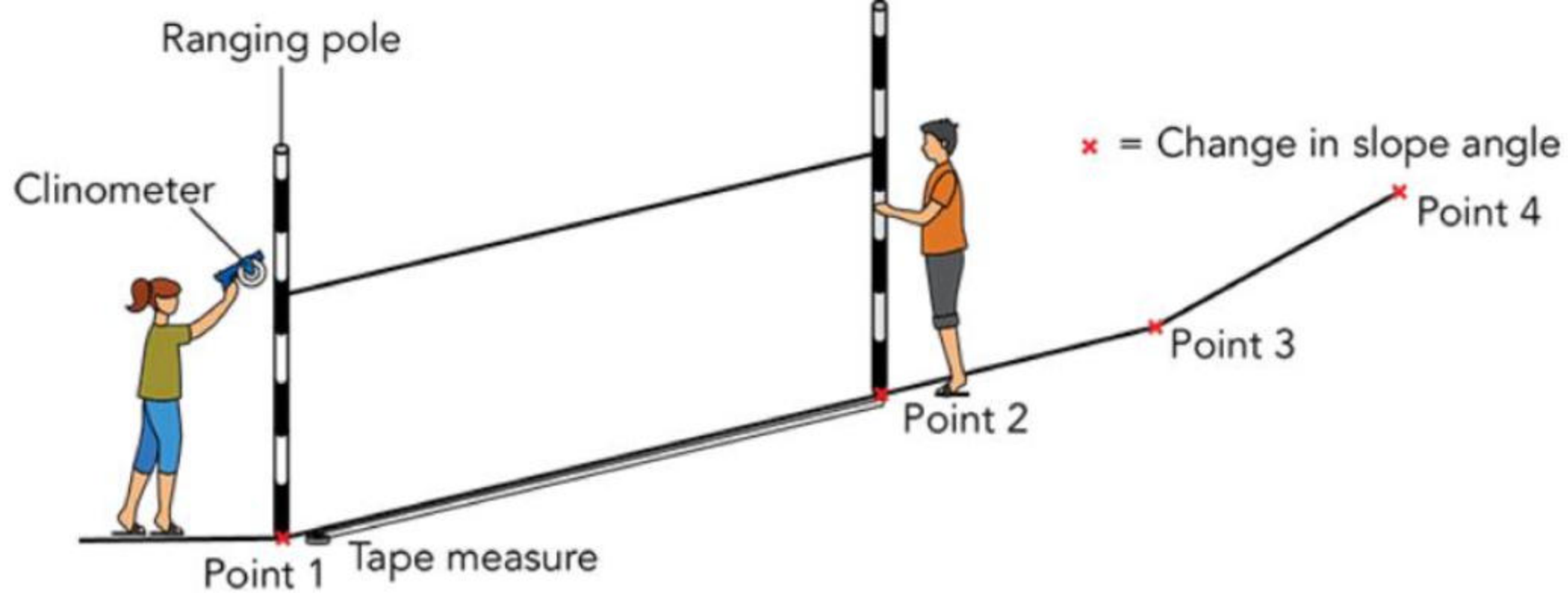


Figure 11.8: The use of ranging poles, a tape measure, and a clinometer to measure the beach profile

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Investigating longshore drift

Analysing swash and backwash can give you an idea of how longshore drift works. You will need an easily visible float. It is best to do this in constructive sea conditions. Where waves are too large and pounding, the float is more difficult to keep track of. Take the following steps:

- 1 Plot a distance of ten metres along a stretch of the beach, marking the start and finish point clearly.
- 2 Place a float at the point at which the waves are breaking.
- 3 Watch the float. Time how long it takes for the float to travel the ten metres that you marked out.
- 4 Alternatively, measure how far the float travels in a given time frame.

TASK 9: Study Source G

Learners collected the data in Source J on the profile of a beach.

- Use this data to plot the three beach profiles.**
- Compare the profiles** at each site. Look for both differences and similarities in the profiles.

Sample site 1		Sample site 2		Sample site 3	
Distance between poles	Angle measured	Distance between poles	Angle measured	Distance between poles	Angle measured
8 metres	6°	4 metres	10°	6 metres	8°
10 metres	4°	8 metres	8°	15 metres	6°
12 metres	2°	15 metres	4°	10 metres	2°
10 metres	4°	12 metres	2°	8 metres	6°
6 metres	2°	6 metres	6°	10 metres	4°

Source J: Profile of a beach

► Image Description

REFLECTION

How did you find combining two different skills to complete this beach profile? Having to think of both working out a scale and creating a graph using a protractor is a new skill. Did you ask for help to get the beach profile angles right when plotting them? Do you feel confident that you now understand this concept?

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Investigating dune vegetation

Investigating changes to vegetation as you move away from the highwater mark can give you insight into how plants colonise dunes and coastal zones. We can

use transects and **quadrats** to study the distribution of organisms in an area along a straight line.

Using a transect and quadrats

A transect is a physical line that is used to analyse a change within a habitat. For example, a piece of string placed along the ground. By analysing the habitat along this line, you can observe and record the habitat that occurs at regular intervals along the transect (Figure 11.9). This data can then be recorded on a graph.

Using a transect for investigating dune vegetation, take the following steps to measure changes in vegetation:

- 1 You will need a pen, paper, table to record data, quadrat, and a long tape measure/rope.
- 2 Working at a right angle to the highwater mark, lay out your tape measure/rope.
- 3 Using the quadrat, at predetermined distances (for example, every five or ten meters), count the number of individual plants and the different number of plant species you find in each quadrat (**Figure 11.9**). Where the plants are too dense to count, you can work out % cover of the plants by estimating how much of the quadrat that plant type covers.
- 4 Record the data for each quadrat, including the distance from the highwater mark each time.
- 5 Analyse the data to determine how vegetation density and diversity changes as you move away from the highwater mark.

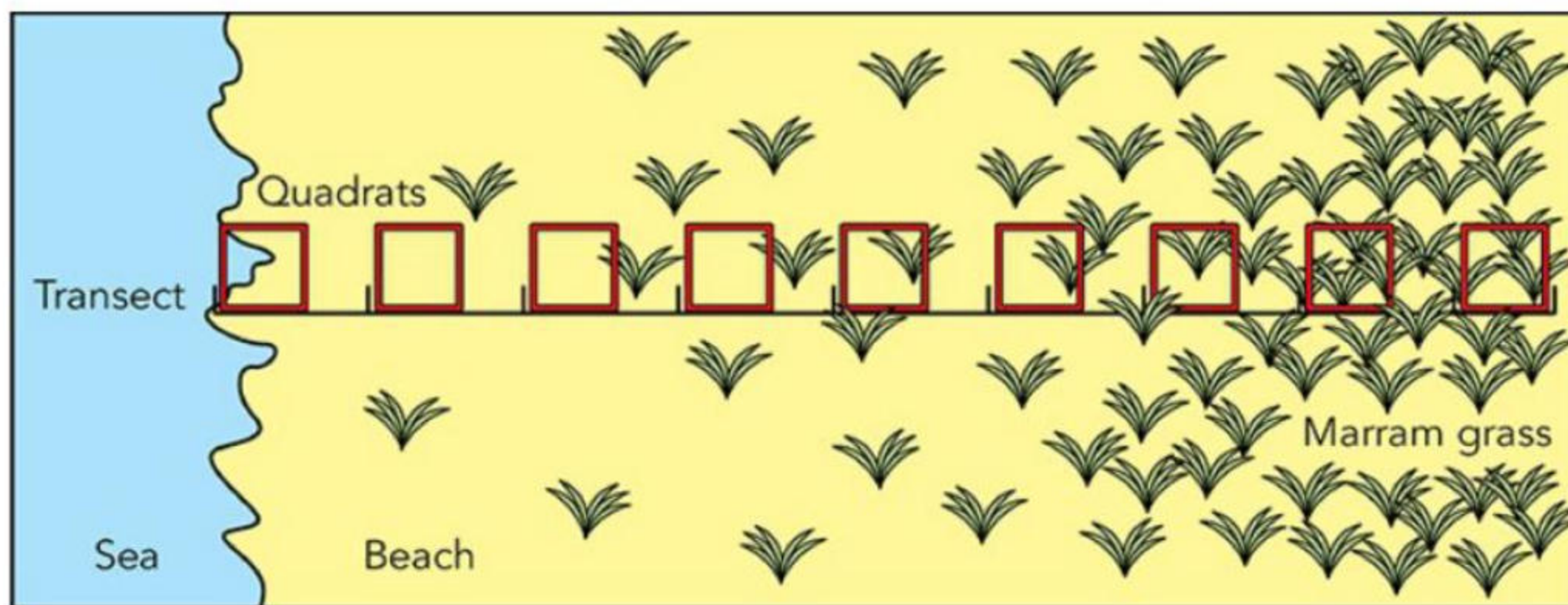


Figure 11.9: A transect running from the highwater mark up to the dunes, showing the placement of quadrats along the transect to sample plant diversity and density

TASK 10

- Design a table** to record the data recorded along a transect 100 metres long, where you are sampling every 10 metres.
- In pairs, discuss how you think vegetation will change as you move away from the highwater mark. Write down two bullet points suggesting how you think it will change.

PROJECT

In pairs or small groups, identify a fieldwork investigation that is related to one of the following categories:

- investigating the CBD
- investigating tourism
- investigating rivers
- investigating a beach profile or a coastline.

Then complete the following tasks:

- Write down an observation.
- Write down a question.
- Write down a clear hypothesis. Remember to make it a statement.
- List three safety precautions you need to take.
- List the equipment you would need for the field trip.
- Explain how you would present your data at the end (charts, graphs, tables).

g Explain why you think your hypothesis would be correct.

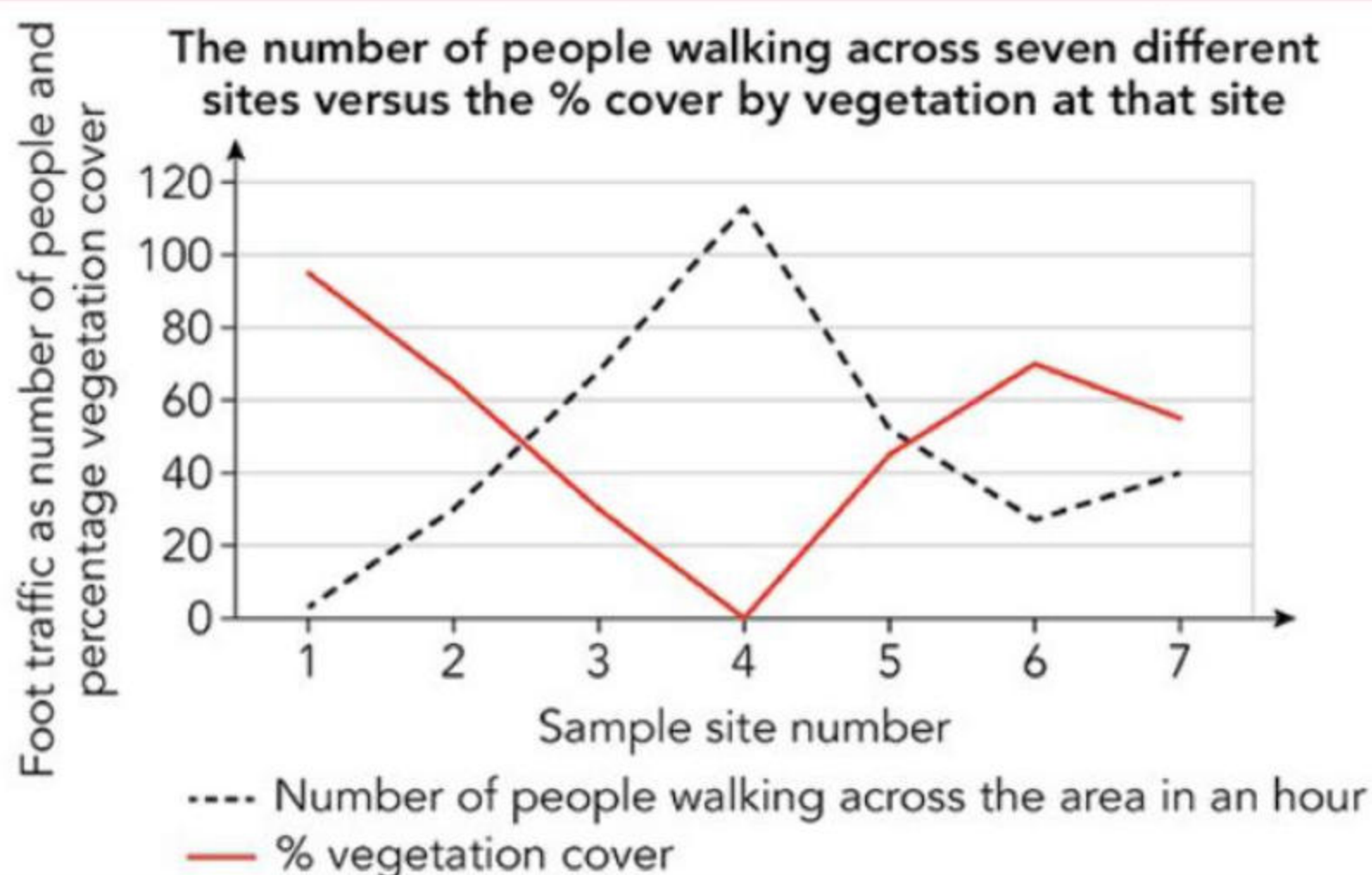
Share your fieldwork idea with the class.

IMPROVE THIS ANSWER

Students investigated the effect of foot traffic on vegetation cover in a field. Their hypothesis was that: Foot traffic will decrease the percentage of vegetation cover found at a site.

The students carried out sampling at seven sites.

Site	1	2	3	4	5	6	7
Number of people walking across the area in an hour	3	30	68	113	52	27	40
% vegetation cover	95	65	30	0	45	70	55



What conclusion would the students make about their hypothesis? Support your answer with evidence from the table and graph given above.

The students tested the hypothesis by checking and recording how many people walked over the seven different sites in an hour. 1 They found that as the number of people walking across an area increased, the vegetation decreased. 2, 3

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1 Although this is correct, it is not what the question is asking. Always ensure you are answering what is being asked.

2 This is correct. However, the answer should include information from the table and the graph to support the answer given. This requires further development.

3 The answer does not state if the hypothesis is true, false, or partially true. It should clearly state the findings.

Rewrite the answer, improving it with the guidance tips given below:

- Read the information off the graph and any supporting data in a table to understand the findings of the investigation.
- Decide if the hypothesis is true, false, or partially true.
- State your decision on the hypothesis. For example, the hypothesis is partially true.
- Support your decision by using data from the graph and the table.

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics.

This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
understand the steps in planning fieldwork using the scientific method			
formulate and evaluate aims and hypotheses			
choose appropriate data collection methods			
consider my safety and required equipment for fieldwork			
make conclusions from the data I collect.			

Preparing for assessment

LEARNING INTENTIONS

By the end of this unit you will be able to:

- understand the assessment requirements for the syllabus,
- understand the assessment objectives knowledge and understanding (AO1), skills and analysis (AO2), and evaluation and decision-making (AO3).
- study more effectively.

The guidance in this unit has been written by the author. References to assessment and/or assessment preparation are the publisher's interpretation of the syllabus requirements and may not fully reflect the approach of Cambridge International Education.

The information in this section is based on the Cambridge International Education syllabus. You should always refer to the appropriate syllabus document for the year of examination to confirm the details and for more information. The syllabus document is available on the website:

www.cambridgeinternational.org.

About the assessment

This syllabus has four components. Students following this syllabus take **three of the four** components:

- Papers 1 and 2 are mandatory
- students will take **either** Component 3 **or** Paper 4.

Paper 1: Physical Geography

This paper is worth 36% of the final grade. It consists of one exam lasting 1 hour and 45 minutes. 75 marks are available.

Paper 2: Human Geography

This paper is worth 36% of the final grade. It consists of one exam lasting 1 hour and 45 minutes. 75 marks are available.

Component 3: Coursework

This component is worth 28% of the final grade. It consists of one assignment of around 2,000 words. 60 marks are available. For O Level, there isn't a coursework option. Instead, students will take the Geographical Investigations paper.

Paper 4: Geographical Investigations

This paper is worth 28% of the final grade. It consists of one exam lasting 1 hour and 30 minutes. 60 marks are available.

Understanding assessment objectives

The assessment for this syllabus is evaluated using three assessment objectives, or AOs. Students need to have a range of skills and so using different assessment objectives helps students to demonstrate more than just a factual recall of information. Understanding assessment objectives will help to improve how you demonstrate your knowledge and understanding, skills and analysis, and evaluative and decision-making skills. The assessment objectives are as follows:

- AO1: Knowledge and understanding
- AO2: Skills and analysis
- AO3: Evaluation and decision-making

What is knowledge and understanding?

Knowledge and understanding is about being aware of and understanding geographical systems, theories, and processes. One example of this could be the river processes of erosion and how they affect people and the environment. You should also consider scale and places, for example, how flooding along the

Mississippi River has local, regional, and national impacts.

What are skills and analysis?

Skills and analysis is about using geographical skills and data This could be presented in many ways, such as graphs, maps, diagrams, and photographs. You may need to interpret and analyse data, such as working out patterns or trends, or you may need to apply data, for example, organising or presenting information.

What is evaluation and decision-making?

Evaluation and decision-making is about evaluating the importance of geographical factors and make well-reasoned decisions. For this, you need to consider the attitudes, values of other people living in different places. You must also consider the impacts of decisions and the need for sustainable development.

The approximate weightings allocated to each of the assessment objectives (AOs) are:

Weighting for assessment objectives

	Paper 1	Paper 2	Component 3 or Paper 4
AO1: Knowledge and understanding	37%	37%	20%
AO2: Skills and analysis	43%	43%	60%
AO3: Evaluation and decision-making	20%	20%	20%
Total	100%	100%	100%

Understanding assessments

Paper 1 and Paper 2

Both Paper 1 and Paper 2 are divided into two sections: Section A and Section B.

In Section A, candidates answer one compulsory structured question.

Question 1 consists of a range of short-answer items and an extended response. Some items are based on source material.

In Section B, candidates answer two out of three structured questions.

Each structured question is worth 25 marks and consists of a range of short-answer items and an extended response. Some items are based on source material.

Paper 4 (Paper 3 for O Level)

This paper contains two compulsory questions each worth 30 marks. Questions consist of a range of short-answer items and an extended response. Some items are based on source material.

The following sections contain practice questions with sample answers written by the author and comments from the author as to why these answers are basic, good and detailed.

Example 1

Question 1: For a named example you have studied, describe the damage caused by an earthquake.

Answer 1: Basic

In 2023, there was a severe earthquake that caused widespread damage and was the deadliest natural disaster in modern history. The earthquake affected a large area and resulted in loss of lives.

Answer 2: Good

On 2 February 2023, a magnitude 7.8 earthquake struck Türkiye and Syria. More than 30 000 aftershocks followed these events. The earthquakes caused damage over an area of approximately 350 000 km², making it the most devastating earthquake Türkiye had experienced in many years. Millions of people were affected and thousands of lives were lost. Millions were left homeless. The most affected regions were in southern Türkiye, where millions of refugees from neighbouring Syria live.

Answer 3: Detailed

On 2 February 2023, Türkiye experienced a magnitude 7.8 earthquake. Nine hours later, a second earthquake occurred, triggering more than 30 000 aftershocks. The impact of these earthquakes was immense, causing widespread damage across an area of approximately 350 000 km². It was the deadliest natural disaster in modern history, leading to the loss of over 55 000 lives, leaving 1.25 million people homeless, and affecting a total of 15.73 million individuals. The World Bank reported that the greatest damage occurred in 11 provinces in southern Türkiye. These regions have high poverty rates and they host more than 1.7 million refugees from neighbouring Syria. This led to a humanitarian disaster.

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Comments

Answer 1 is a basic answer. It doesn't name a specific earthquake, but it does give a date. There are no details about the damage caused by the earthquake, which means the question hasn't really been answered. The question asked about a named earthquake, but this answer could refer to many different earthquakes.

Answer 2 names a specific earthquake. It says how big the earthquake was and when it took place. There is some detail to the answer, such as the area damaged by the earthquake. This answer also recognises the scale of the earthquake by pointing out that it was one of the most devastating in Türkiye's history.

Answer 3 has much more detail. It names the earthquake and give lots of specific detail about when it took place and how large it was. The description of damage is very detailed, giving many facts about how many people and how much land was affected. Statements are supported with detail about where the information was derived from. The answer fully addresses the question.

Example 2

Question 2: For a country you have studied, describe the causes of food insecurity.

Answer 1: Basic

Haiti faces food insecurity because of things like natural disasters and dependence on food imports.

Answer 2: Good

Much of Haiti's food insecurity is rooted in history. The country has been politically unstable, and it experienced a political coup. It has also had increasing crime rates and natural disasters, which damage infrastructure and cause diseases like cholera. Frequent hurricanes bring strong winds and flooding, which destroys crops. Drought also causes crop failure.

Answer 3: Detailed

Haiti's food insecurity can be attributed to social, environmental, and economic factors. Historically, the country has been marked by political instability and has had the lowest Human Development Index (HDI) ranking in the Americas. A political coup, ongoing economic deterioration, and rising crime rates have severely limited the country's ability to produce and distribute food to its population. The catastrophic earthquake of 2010 led to approximately 250 000 deaths and damaged critical infrastructure, transport routes, and food supply chains. A cholera outbreak in the earthquake's aftermath then weakened the workforce, leading to a reduction in overall food production. In addition to this, Haiti is a hurricane-prone area of the world. Hurricanes bring strong winds and flooding, destroying crops and essential infrastructure. Haiti also regularly suffers from prolonged periods of drought and poor soil quality, which can limit crop growth, exacerbating the food security challenge. Importantly, over half of the food and 83% of the rice consumed in Haiti are imported, meaning that the country is susceptible to fluctuations in international food prices.

Comments

Answer 1 is a basic answer. It names a country and it answers the question by giving two causes of food insecurity. The answer doesn't give any detail, for example, what natural disasters take place in Haiti? The answer also only mentions two causes of food insecurity, and adding some more causes would give more detail and depth to the answer.

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Answer 2 names a country and gives several causes of food insecurity. The use of the word 'which' shows that the points are explained in the answer. It would be better if these explanations linked with the question more clearly, for example, the answer states that natural disasters 'damage infrastructure and cause disease like cholera' but it doesn't explain how that links to food insecurity.

Answer 3 has much more detail. It names the country and gives lots of specific detail about why it faces food insecurity. Descriptions are very detailed, giving many facts such as 83% of rice consumed in Haiti is imported. Points are well developed. For example, it explains why a cholera outbreak led to a reduction in food security. The question is fully answered.

Example 3

Question 3: For a coastline you have studied, describe the impacts of erosion.

Answer 1: Basic

The impacts of coastal erosion include the loss of beaches, damage to property and damage to habitats.

Answer 2: Good

Coastal erosion in Florida has widespread impacts, including the loss of beaches, damage to property and the destruction of habitats. Issues such as damage to property cost a lot of money to repair and rebuild, which affects the economy. Mangroves are also damaged, which is bad for the environment

Answer 3: Detailed

The coastline of Florida has been badly impacted by coastal erosion with 696 km of Florida's beaches being classified as critically eroded. In Miami Beach and Fort Lauderdale, erosion has led to the gradual disappearance of sandy beaches, which are important for tourism. This has led to expensive beach renourishment projects to protect tourism and protect coastal infrastructure. In places like Naples and Sarasota, erosion has caused damage to waterfront properties, such as homes and hotels. Additionally, erosion poses a big threat to Florida’s natural habitats, such as the Everglades and Florida Keys, where mangroves are being lost.

Comments

Answer 1 is a basic answer. It does not name a coastline. It answers the question by giving three impacts of erosion, but it does not give any detail about each impact, for example, how does the loss of beaches affect the area? Adding more detail and depth would improve this answer.

Answer 2 names a stretch of coastline and gives several impacts of coastal erosion. The answer explains several points, for example that damage to property costs a lot of money to repair and rebuild. There are some good key terms used, such as habitats and economy. This answer would be better if it located the impacts, for example, where are the beaches located?

Answer 3 has much more detail. It names an area of coastline and gives lots of specific detail about the consequences of erosion. Descriptions are very detailed, giving the locations for the different impacts. Points are well developed, for example that the loss of beaches would affect tourism and lead to costly renourishment projects. This answer looks at the impacts in four named locations along the Florida coastline, which gives it depth and detail.

Skills-based questions

Example 1

Question 1: Using Figure 1, describe the change in population in Singapore between 2000 and 2020. [3]

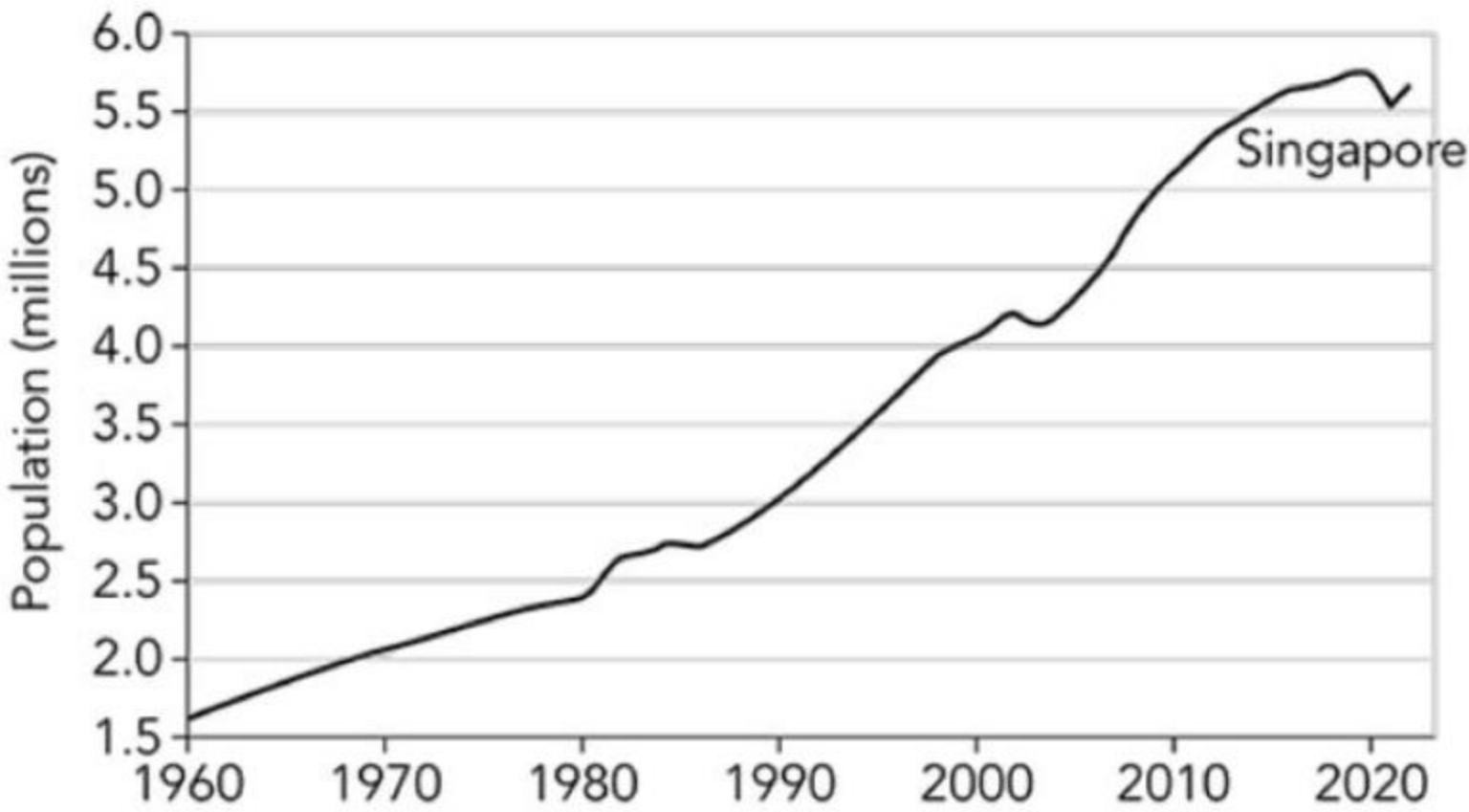


Figure 1: Total population of Singapore, 1960–2023.

Answer 1: Basic

Singapore’s population has increased.

Answer 2: Good

Singapore’s population has steadily increased between 2000 and 2023. The population also decreased twice during this time.

Answer 3: Detailed

Singapore’s population has steadily increased from around 4 million people in 2000 to around 5.6 million people in 2023. During this time, there have also been two decreases in population; these took place between 2002 and 2003, and also between 2019 and 2021.

Comment

Answer 1 is very basic. It gives an overall trend but does not provide any detail. Answer 2 gives a little more detail. It gives a time frame, and also recognises that the population has also declined twice. Answer 3 is much more detailed. It uses specific facts from the graph to estimate the total population at given times. It also gives years for the anomalies, where the population has decreased, rather than increased.

Example 2

Question 2: Describe the distribution of species along the transect in Figure 2. [3]

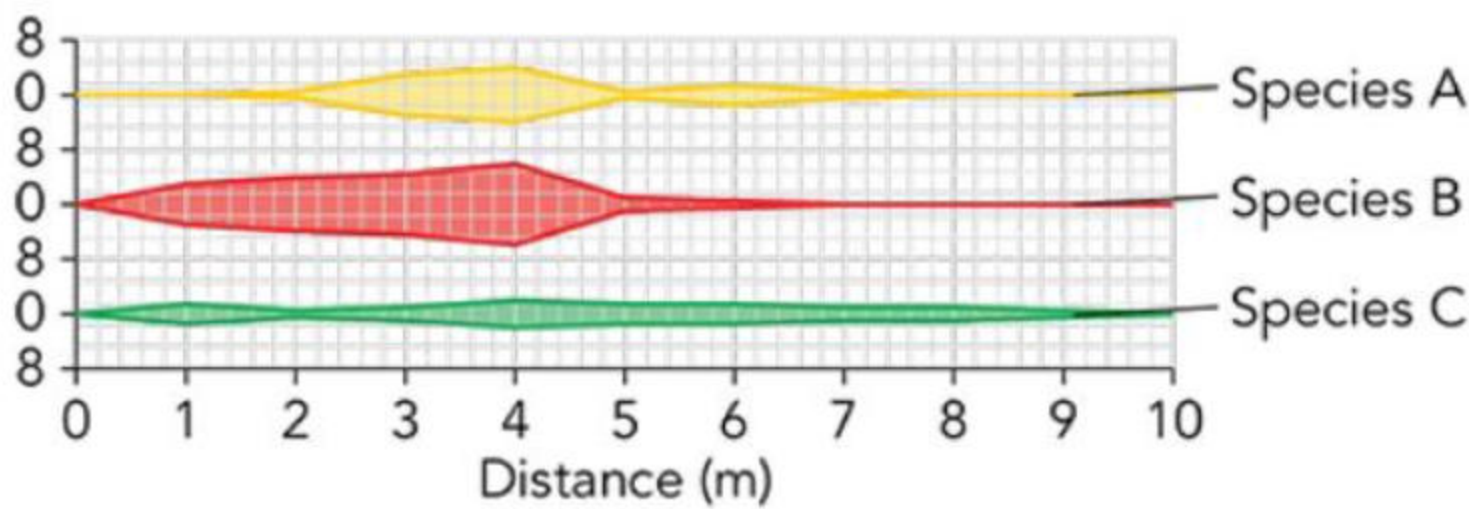


Figure 2: Distribution of species along a transect

Answer 1: Basic

Species A, B, and C are not evenly distributed along the transect.

Answer 2: Good

Species A, B, and C are not evenly distributed along the transect. B is the most common species and is found nearer the start of the transect.

Answer 3: Detailed

Species A, B, and C are not evenly distributed along the transect. A and B are more common at the start of the transect than the end. Species C is quite evenly spread throughout the transect. No species were recorded at the end of the transect.

Comment

Answer 1 is very basic. It gives an overall trend but does not provide any detail, such as how the species are spread. Answer 2 gives more detail, in that it names species by letter and says which is the most common. Answer 3 is much more detailed. It gives an overview and then describes the spread of each of the three species. It also identifies a point in the transect where there were no recordings of species A, B, and C.

Example 3

Question 3: Study Figure 3, which shows earthquake damage in Khokana village, Nepal. Describe the damage seen in the photograph. [4]



Figure 3: Earthquake damage in Khokana village, Nepal

Answer 1: Basic

Houses are damaged and there is rubbish everywhere. People will not be happy.

Answer 2: Good

The roads are blocked with debris, which may have fallen from the damaged buildings.

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Answer 3: Detailed

There is damage to the buildings, as shown by the corrugated iron and the debris in the road. The road is blocked with rubble. There are power lines falling down, which shows that the electricity supply may have been cut off.

Comment

Answer 1 is very basic. It gives an overall judgement, but it also makes an assumption that ‘people will not be happy’. As this is something that cannot be seen in the photograph, this is not a suitable response. Answer 2 gives a little more detail, in that it gives two types of damage. Answer 2 also uses the key term ‘debris’. Answer 3 is much more detailed. It identifies three types of damage: buildings, roads, and electricity cables.

Example 4

Question 4: Food supplies can be made more sustainable. Figure 4 below describes some of these methods.

A shift towards a more sustainable global food system could create up to \$10tn (£7.9tn) of benefits a year, improve human health and ease the climate crisis, according to the most comprehensive economic study of its type.

It found that existing food systems destroyed more value than they created due to hidden environmental and medical costs, in effect, borrowing from the future to take profits today.

Food systems drive a third of global greenhouse gas emissions, putting the world on course for 2.7°C of warming by the end of the century. This creates a vicious cycle, as higher temperatures bring more extreme weather and greater damage to harvests.

Figure 4: Extract from the Guardian Newspaper, January 2024

Using information from Figure 4 only, describe:

Two benefits of moving towards a more sustainable global food system.

Two problems with existing food systems.

Answer 1: Basic

Sustainable food is better, but existing food systems create greenhouse gas emissions.

Answer 2: Good

Sustainable food is beneficial because it helps to ease the climate crisis by reducing emissions. Existing food systems drive a third of greenhouse gas emissions, which brings extreme weather.

Answer 3: Detailed

Sustainable food is beneficial because it improves human health and it helps to ease the climate crisis because it produces fewer greenhouse gas emissions. Existing food systems have hidden environmental and medical costs, such as the use of intensive farming methods and pollution. Existing systems also drive a third of greenhouse gas emissions, which are contributing towards climate change.

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Comment

Answer 1 is very basic. It gives one problem with existing food systems, but it doesn’t explain why sustainable food is better, it just states that it is better. Answer 2 gives a little more detail, in that it gives one benefit and one problem. Answer 2 also makes a clear link between greenhouse gas emissions and extreme weather. Answer 3 is much more detailed. It identifies two benefits and two problems, so it fully answers the question. It also explains the text, rather than just ‘lifting’ it from the Figure.

Geographical Investigations

Example 1

Question 1: Students were learning about processes taking place along a coastline.

Use arrows to match the processes with the correct definitions in Table 1 below. [3]

One has been completed for you.

Process
Corrosion
Hydraulic action
Attrition
Corrasion

Definition
The slightly acidic nature of sea water breaks down material. Chalk and limestone cliffs are particularly prone to this.
Material carried by the sea rubs against cliffs, wearing them away.
Water is forced into cracks in cliffs which compresses the air inside. This force breaks off bits of rock
Material carried by the sea, such as pebbles, collide with one another. This wears them down to form smaller, rounder particles.

Table 1: Coastal processes and definitions

Comment

This question asks for arrows to be drawn from the name of the process to the correct definition. It can be easy to not see these types of questions that don’t have answer lines, so it’s important to take care when answering all questions.

Example 2

Question 2:

Table 1 shows processes of erosion. Suggest two other coastal processes. [2]

Answer 1

Erosion

Answer 2

Things being moved

Things being deposited

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Answer 3

Transportation

Deposition

Comment

It is important to notice the word ‘other’ in the question. The answers should therefore not include ‘erosion’. Answer 1 is therefore incorrect. Answer 2 gives the correct descriptions of other processes, but it does not give the correct names for the processes. Answer 3 gives two other processes and accurately names them.

Example 3

Question 3:

Students carried out some fieldwork to measure the velocity of a river at different locations. Describe how they would measure the velocity using the following equipment:

- two ranging poles
- a tape measure
- a stopwatch
- an object that floats, like a ball or a cork.

[4]

Answer 1

You could use the stopwatch to measure how long it takes for the ball to float along a river.

Answer 2

You could measure out a section of river and throw the ball into the river at the start of the measured area. You can then time how long it takes for a ball to float along it. You could then calculate the speed of the ball.

Answer 3

You could measure out a section of the river with the tape measure. Each end would then be marked using a ranging pole. The floating object would be placed at the ranging pole upstream and, when it is put into the water, the timer is started. As the ball passes the second ranging pole, the timer is stopped. From this, you can calculate the speed, or velocity, of the ball. The ball moves as fast as the river, so this would give the velocity of the river too.

Comments

The first answer is accurate, but it does not give any detail about how the technique should be carried out. The second answer is better as it gives a little more information about measuring out the river. However, it suggests that the students ‘throw’ the ball into the river, which could lead to inaccurate results. Answer 3 gives a detailed overview of the technique. It uses all four pieces of equipment, and it links the speed of the floating object to the speed of the water.

Example 4

Question 4:

The data collected is shown in the table below:

Table 2: Data collected about the speed of flow of the river

	Time to travel 10 metres	Velocity
Site 1 (upstream)	114	0.08 m/s
Site 2	98	
Site 3	82	0.12 m/s
Site 4	140	0.07 m/s
Site 5	79	0.13 m/s
Site 6 (downstream)	68	0.14 m/s

The method used to calculate the velocity of the river at each site is shown below.

- i Complete Table 2 by calculating the velocity at site 2.

[1]

Comment

This question asks for the table to be completed, using the formula above. The answer should be written in the same way as the others in the table, which means using two decimal places and writing m/s after the figure. In this case, the correct answer should be 0.10 m/s.

- ii Suggest reasons why the river at site 4 is flowing much more slowly than expected.

[2]

Answer 1

The river here is much slower than at the other sites.

Answer 2

The river at site 4 could be flowing over flatter ground, which would make it slower.

Answer 3

Site 4 could have a gentler gradient, which would make the river slow down. The slower speed could also be down to problems with the data collection, for example, if the ball had got stuck behind rocks, then the river would appear to be flowing more slowly.

Comment

Answer 1 is far too basic and it doesn't answer the question. Answer 2 gives a good suggestion about why the velocity is slower at site 4, but the answer is very brief. Answer 3 gives two good suggestions about why site 4 has a slower velocity, both of which are well explained.

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Example 5

Question 5:

Students carried out some fieldwork to investigate depth and width along a section of a river.

They tested the following hypothesis: 'as river width increases, river depth also increases'.

To test this hypothesis, they collected information about the width and depth of a river at six different sites. Their data is plotted in Figure 1 below.

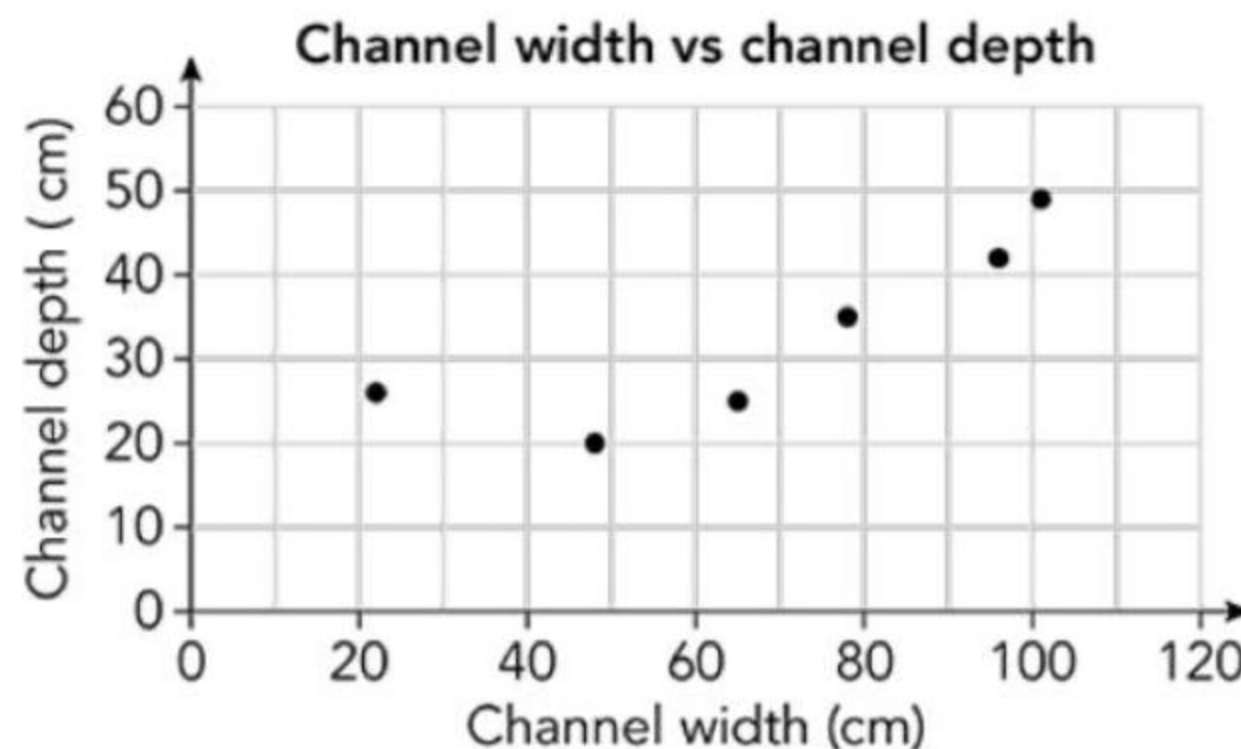


Figure 1: Data collected about the width and depth of a river

► **Image Description**

Using Figure 1, decide whether the student's hypothesis was correct. Use data to justify your decision

[6]

Answer 1

The graph shows that the hypothesis is correct as it shows how width and depth changes over the river.

Answer 2

The graph shows that depth increases as width also increases, so the hypothesis is correct. The widest section of the river is also the deepest section. It is over 1m wide and around 50cm deep at this point.

Answer 3

The graph shows that depth increase as width increases. The deepest part of the river is around 50 cm deep and 1m wide, and the shallowest part of the river is around 20 cm deep and just under 50 cm wide. However, this hypothesis is only partially correct because the relationship is not perfect. One of the sites has a very narrow width of around 22 cm, but the river here is quite deep, at around 26cm. The general trend is still a positive correlation, whereby depth and width both increases, but there is at least one anomaly.

Comments

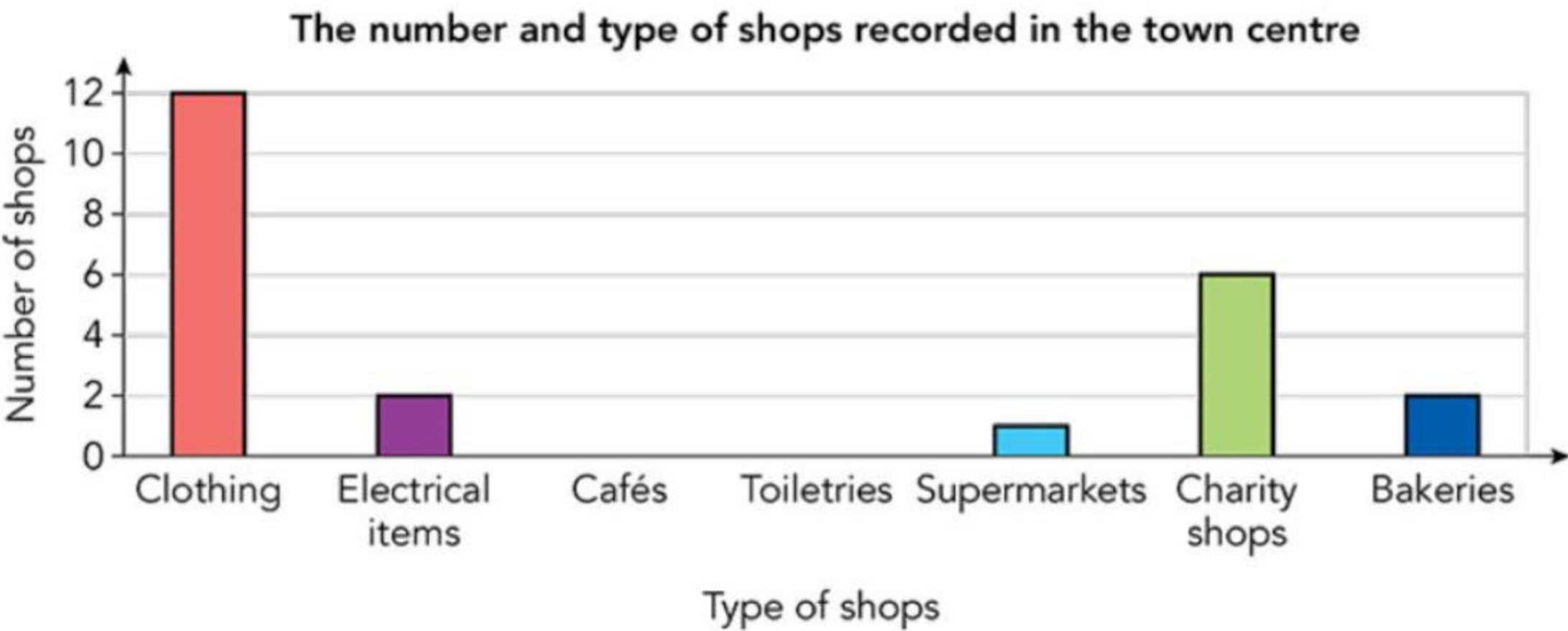
Answer 1 gives very little detail. It shows a reworded version of the text above and does not say why students made the conclusion that width and depth are related. Answer 2 is better as the relationship between width and depth of the river is clearly stated, and specific data has been given. However, it does not recognise that the data isn't a perfect correlation, which would mean that the hypothesis is only partially correct. Answer 3 is better because it gives much more detail. It provides a detailed overview of the relationship between width and depth of the river, plus it acknowledges the anomaly that exists with the data collected at the narrowest site.

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Example 6

Question 6:

Figure 2 shows data collected about a shopping centre. The table below shows the data that was used to create the graph.



Shop type	Number of shops
Clothing	12
Electrical items	2
Cafés	7
Toiletries	5
Supermarkets	1
Charity shops	6
Bakeries	2

Figure 2: The number and type of shops recorded in the town centre

Complete Graph 2 (above), to show the number of cafés and toiletries shops. [2]

Comment

The types of questions that asks for graphs to be completed are very easy to miss. The question requires plotting of bars to show information from the table below. Lines should be drawn carefully and with a ruler.

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Example 7

Question 7:

The students who collected the data for graph 2 wanted to find out what people thought about the mix of shops in their town centre. When their teacher approved the questionnaire, she advised students how to approach people who were shopping. Suggest three pieces of advice which she might have given them. [3]

Answer 1

- 1
- Tell people that they have to help you.
- 2
-
- 3
-

Answer 2

- 1 Don't block pavements.
- 2 Ask at least two people.
- 3 Go in the evening, when it is less busy and people have more time.

Answer 3

- 1 Work in pairs and not alone.
- 2 Be polite and say thank you.
- 3 Ask a range of different people.

Comments

Answer 1 only gives one suggestion. The suggestion is also incorrect, in that people do not have to respond to questionnaires. Answer 2 makes three suggestions. The first suggestion is useful, but the second and third are less helpful because questionnaires need to ask a wide range of people, not just two, and because it's better to choose a time when the centre is busy. Answer 3 suggests three useful pieces of advice.

Example 8

Question 8:

To extend the town centre study, students compared the current shop types with the numbers of shop types recorded on an old map of the town centre. Which one of the following is the correct description of the use of old maps. Tick your choice. [1]

	Tick
Primary data	
Secondary data	
Qualitative data	

Comment

The use of old maps is secondary data. Primary data is information that you have collected yourself. Qualitative data is information presented as words or pictures and this data is numeric.

Example 9

Question 9:

The students carrying out the questionnaire in the town centre used the following method to choose which shoppers to talk to:

‘Give the questionnaire to older women because they know the town centre better than younger people or men’.

Explain why this is a poor method to obtain a representative sample. [3]

Answer 1

You might not be able to find any older women whilst you're there.

Answer 2

It is important to ask men as well as women. Men might use different types of shops.

Answer 3

It is important to survey different types of people so that you get a full picture. This means interviewing men and women, and also finding people that fit into different age groups.

Comment

Answer 1 is very basic and is not a reason why the sampling technique is inaccurate. Answer 2 is accurate, but it fails to recognise the need to interview people of different ages. Answer 3 identifies two issues with the sampling technique and explains why it is important to survey a range of different types of people.

Example 10

Question 10:

Students carried out surveys about the impact of a ski resort on a local town. A questionnaire identified seasonal employment as a problem of tourism.

Answer 1

Seasonal employment is bad for tourist resorts because seasonal jobs are not good.

Answer 2

Seasonal employment is a problem because it means that people only have jobs for certain times of the year.

Answer 3

Seasonal employment means that people do not work for the full year. This means they must find other jobs out of season. Seasonal work can also be lower paid than other types of work, which means that people have less money to spend.

Comments

Answer 1 is basic; it recognises that seasonal jobs can be bad, but it doesn’t explain how or why. Answer 2 explains why seasonal jobs can be bad, but the explanation is very brief. Answer 3 is much more detailed, giving two reasons why seasonal employment can be bad, and explaining each.

Example 11

Question 11:

Tourists at the ski resort also complained about an increase in traffic during the skiing season. As an extension to this study, students decided to do traffic counts. They chose a day during the skiing season, when more tourists were in the town.

The results of the traffic survey are shown in Table 4. Describe the method that the students used to gather this information.

[4]

Table 4: Results of traffic count at one site

Method of travel	Time of traffic count (counting for one minute in both directions)					
	09.00	11.00	13.00	15.00	17.00	19.00
Car	14	21	13	16	32	22
Tourist coach	8	5	4	3	7	1
Local bus	2	3	2	2	3	2
Taxi	1	2	1	2	3	3
Bike or motorbike	3	4	3	6	2	3

Answer 1

Students visited the site at the different times of the day. They counted how many vehicles they could see.

Answer 2

Students visited the site at six different times during the day. They did a tally to show how many vehicles they could see in one minute in both directions.

Answer 3

Students visited the site at six different times of the day, every two hours between 09.00 and 19.00. They used a timer to time one minute, and during that minute they counted the number of vehicles that passed by in one minute. This needed more than one person because someone needed to use the timer and two people needed to count the different directions.

Comment

Answer 1 gives a brief overview of the technique used to count traffic. It does not give any detail about timings for either visiting the site or counting the traffic. Answer 2 recognises the number of visits to the site and that the vehicles were counted in both directions. It suggests the vehicles were done using a tally, which isn’t evident from the table, but which would have been likely. Answer 3 gives much more detail, in that it describes the spacing between the visits and the start and end time for the day. It also describes how the activity would be timed and how it might require more than one person.

Example 12

Question 12:

To find out more about what locals thought about the ski resort, the students asked Question 3 in the questionnaire:

‘What do you think are the benefits and problems of tourists visiting this area?’The results are shown in Table 5.

Using the data in Tables 4 and 5, to what extent does tourism benefit local people? Include information from Tables 4 and 5 to justify your answer. [6]

Table 5: Benefits and problems of tourists visiting a ski resort

Benefits	Problems
More money spent in local shops and restaurants	Congestion is very bad, particularly when the large tourist buses block the narrow roads
More jobs in the local area	Some people have bought second homes here, which makes property prices more expensive
We love to ski, so having the resort on our doorstep is great	Money spent in shops and restaurants is seasonal
The fact that the ski resort is here means that we’re lucky to live in a beautiful place	There are lots of strangers in our town
Public transport is better since the tourists have arrived	The types of jobs created by the ski resort are seasonal and low paid
The town is busy – without the skiing it would be very quiet	Some people behave badly when on holiday

Answer 1

Tourism benefits locals in the ski resort because it provides jobs and brings money into the area. This is good because it means people have more money to spend. The town is also much busier, which some people like.

Answer 2

Tourism is both good and bad. It brings money into the local area and the presence of tourists means that money has been invested into public transport. It also makes the place more beautiful. However, it is also bad because jobs are often seasonal and property prices have increased because of second home ownership. This means it can be both.

Answer 3

Tourism benefits locals to some extent, but it also brings some problems. It is beneficial because it provides jobs and brings money into the local economy. The fact that tourists visit means that money is also invested into the local infrastructure, providing services like buses. However, tourists also bring many problems. These include an increase in property prices and low-paid, seasonal employment, which when combined can price locals out of the area. Table 4 also shows that there are problems with congestion, particularly with large tourist buses dropping off and picking up tourists. A total of 28 tourist coaches were counted in just one day, and these were often arriving and leaving at the same time – around 9 am and 5 pm. Overall, whilst tourism benefits locals to some extent, it also brings many problems.

Comment

Answer 1 is biased, in that it only looks at the benefits of the resort. The question asked about the extent to which tourists bring benefits, so both benefits and problems need to be considered. Answer 1 also does not use Table 4. Answer 2 looks at both benefits and problems and recognises that there are both. However, it also does not use Table 4. Another issue with answer 2 is that it shows a misunderstanding of one of the answers in the questionnaire; the respondent said the presence of tourists shows that they live in a beautiful place, and not that tourism ‘makes the place more beautiful’. Answer 3 is better, in that it considers both tables and clearly explains to what extent tourism is beneficial. Answer 3 also uses data from Table 4, because it adds up the number of tourist coaches and links this to the question.

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